Learning to Schedule: A Twelve-Week Policy Discovery Experiment

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UCSC Graduate Capstone Stack: Python 3.10+, pure NumPy + stdlib (no PyTorch / TensorFlow)

Abstract

We report a twelve-week experiment in tabula rasa reinforcement learning for CPU scheduling on a 5-process stochastic environment, with no hardcoded policy, no expert demonstrations, and no handcrafted features. The agent learns from a dense reward signal and is evaluated against a Shortest Remaining Processing Time (SRPT) oracle and a Round Robin baseline.

The progression exposed nine distinct failure modes and their resolutions: (Week 1) a tabular Q-learning agent beat Round Robin but produced zero cross-episode variance, revealing a discretisation ceiling where identical Q-values were assigned to all burst lengths within the same bin; (Week 2) randomised training extended state coverage to 5,386 states (32% of the discrete space) but could not resolve the ceiling because it was structural, not a coverage problem; (Week 3) a numpy-only DQN with He initialisation, experience replay, and a target network escaped the discretisation ceiling — std = 26.51ms across 500 random episodes, within 2.55ms of the SRPT oracle's natural baseline variance — but position-indexed action heads generated identity bias (|Δ| = 5.48, wrong direction) and a load-adaptive quantum pattern that appeared to encode a learned heuristic; (Week 4) policy extraction from 10,000 greedy decisions identified that the quantum pattern was coherent, reproducible, and load-correlated, but diagnosably PID-positional in origin; (Week 5) an action-conditioned architecture (19→64→32→1) with candidate-slot masking achieved partial permutation invariance, reduced mean MCT from 85.00ms to 76.04ms (−10.5%), narrowed the SRPT gap from 17.34ms to 8.38ms, resolved identity bias to |Δ| = 0.051 in the correct direction, and eliminated the spurious quantum pattern entirely; (Week 7) an AttentionDQN replaced the flat competitor concatenation with dot-product attention over competitor key-value pairs, achieving full permutation invariance over the competitor context — but attention entropy collapsed to H = 0.03 without regularisation, entropy annealing (λ: 0.10 → 0.005) was required to stabilise learning, and the resulting agent trades higher mean MCT (80.57ms, +12.91ms vs SRPT) for lower variance than the SRPT oracle itself (std = 21.82ms vs 23.96ms), meeting the 80% SRPT-agreement target at n_active ≤ 3 but failing at n_active = 5 due to residual attention entropy; (Weeks 8–10) real-trace transfer required outlier filtering (p75 cutoff), log-normalisation, and resource-class features (plan_cpu, plan_mem) to beat Round Robin on Alibaba 2018 batch_task data for the first time — W10C two-head attention achieved mean MCT 18.21s, beating Round Robin by 3.10s with spontaneous head specialisation into load-context and competitor-monitoring roles; (Week 11, W10C-VC1) replacing the area-integral reward with a value-curve objective without making curve parameters (tau, floor) observable produced non-stationary Q-values — the identity bias probe returned the same Δ = +0.035765 across all four curve-type combinations, confirming the network was blind to the reward's defining features and could not converge; (Week 12, W10C-VC2) extending the per-process feature vector to include tau_norm and floor produced urgency-differentiated Q-values and a widening steep/smooth SRPT agreement gap (9.7pp at best checkpoint), confirming that reward signal design determines which structural behaviours are learnable, but at measurable MCT cost (best MCT 21.46s vs W10C 18.21s) because value preservation and mean completion time structurally diverge for slow-decaying processes — a tension confirmed at the oracle level before any learning occurs.

The reward-objective tension between value preservation and MCT — and the finding that a perfectly greedy value-curve oracle cannot beat Round Robin on MCT — identifies reward signal design as the primary remaining open problem in extending tabula rasa RL scheduling to multi-objective workloads.

Three formal results accompany the experimental progression. First, a proof-of-monotonicity theorem (Section 3) establishes that per-task quantum scoring is structurally degenerate for value-curve schedulers: any rate-normalised scorer always selects the minimum quantum; any total-value scorer always selects the maximum. The Corollary (Section 3.1.3) identifies queue-global information as a necessary condition for non-degenerate quantum selection, confirmed by three ablations in Section 5.23. Second, a calibration study (Section 5.26) shows that τ values calibrated to trace-scale delays eliminate the zero-gradient problem but still produce near-ties in 93–100% of ordering decisions on batch-dominated workloads, because the near-tie condition is trace-composition-determined. Third, we introduce the Value-Rate Fairness Index (VRFI = 1 − CV(VLR), Section 5.27.2), a metric sensitive to value-curve heterogeneity that Jain's Fairness Index and slowdown variance cannot detect; value_aware achieves rank sum 17 with zero starvation across five evaluated policies, and VRFI = +0.289 under fairness-targeted parameter optimization.

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1. Introduction

Modern CPU schedulers operate on what is immediately observable: the processes in the ready queue, their recent CPU usage, and their priority class. Linux's Completely Fair Scheduler (CFS) distributes CPU time proportionally to virtual runtime, maintaining fairness over runnable tasks. But fairness over what is visible is not the same as minimising the time tasks spend waiting to become visible. A short-burst process that arrives while the CPU is occupied by a long-burst process experiences a waiting time that CFS cannot reduce by design — the problem is not in its policy but in its horizon.

The question we ask is different: can an agent, given a complete view of each process's remaining burst and arrival state, learn a scheduling policy from scratch — no supervision, no target policy to imitate, no handcrafted priority rules — purely from the signal of how long processes have been alive? This is a policy discovery problem, not a CFS improvement problem. We do not propose to replace CFS. We ask whether RL can, in a controlled toy environment, rediscover approximations to the known optimal policy, and — more importantly — whether the failure modes that prevent full rediscovery reveal something about the architecture requirements for generalising scheduling policies across arbitrary workloads.

Our contribution is a seven-agent progression with honest reporting of both the positive results and the failures:

1. A tabular Q-learning agent that beats Round Robin but hits a discretisation ceiling (zero cross-episode variance, identity bias pointing in the wrong direction).
2. A randomised-training variant that extends state coverage to 32% of the discrete space but cannot shrink the identity bias below the bin-resolution threshold — a structural, not sampling, limit.
3. A numpy-only DQN that unlocks continuous state and non-zero variance, but whose position-indexed action heads introduce a new failure mode: identity bias worsens from $|\Delta| = 0.225$ to 5.48, and a load-adaptive quantum pattern emerges that appears to be a learned heuristic.
4. A policy extraction experiment that identifies the quantum pattern as a PID-positional artefact masquerading as load-adaptive behaviour.
5. An action-conditioned DQN that achieves partial permutation invariance, resolves identity bias, eliminates the spurious quantum pattern, and closes the SRPT gap from 17.34ms to 8.38ms.
6. An AttentionDQN (Week 7) that achieves full permutation invariance over the competitor context via dot-product attention, but collapses without entropy regularisation and requires an annealed entropy penalty (λ:

0.10 → 0.005) to thread the needle between softmax freeze and TD-signal domination — meeting the 80% SRPT-agreement target at n_active ≤ 3 while failing at n_active = 5 due to residual entropy near termination.

7. A joint diagnostic that resolves an apparent contradiction — 80.2% attention-on-longest while agreeing with SRPT 82.2% of the time — by showing that attention tracks the most threatening competitor (longest remaining burst) as structural context for correct short-process selections.

8. A formal monotonicity proof (Section 3.1.3) showing that per-task quantum scoring is structurally degenerate for value-curve schedulers: f_rate always selects q_min, f_total always selects q_max. The Corollary establishes that queue-global information is a necessary condition for non-degenerate quantum selection in any value-preserving scheduler.

9. Three ablation experiments (Section 5.23) confirming the Lemma: quantum_only (f_rate) selects q_min 100% of the time; quantum_only_fixed (f_total) selects q_max 100%; ordering_only with fixed quantum collapses to near-zero total value (15.37) through smooth-task starvation.

10. A tau calibration study (Section 5.26) identifying the disconnect between default τ values (25s/100s) and trace-scale delays (medians 5,279s/38,665s), and showing that calibrated τ values (900s/800s) still produce near-ties in 93–100% of ordering decisions because the near-tie condition is trace-composition-determined, not τ-determined.

11. The Value-Rate Fairness Index (VRFI, Section 5.27.2), a new metric defined as 1 − CV(VLR) that captures value-curve fairness invisible to Jain's FI and slowdown variance — demonstrated by a key falsification example where Jain's = 1.0, SDV = 0, yet VRFI = 0.641.

12. A five-policy fairness analysis (Section 5.27.3) showing value_aware achieves rank sum 17 with zero starvation, SDV = 566 (28× lower than Round Robin), and confirms that global VRFI is negative for all preemptive policies on mixed steep/smooth workloads due to structural base/τ divergence between task types.

13. A gradient descent analysis of value curve parameters (Section 5.28) exposing a three-level degeneracy hierarchy under efficiency loss (floor collapse → base shrink → tau drift) and showing that VRFI-based optimization is the only exploit-resistant objective, achieving the first positive VRFI (+0.289) on this trace.

The honest negative results — zero variance as a sign of discretisation collapse, identity bias worsening with the move to DQN, the 80% process-selection target not met — are as central to the contribution as the positive ones. Each failure is reproducible, diagnosed to a specific architectural cause, and fixed or explicitly left open.

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2. Background

2.1 CFS and the Scheduling Horizon Problem

CFS uses a red-black tree ordered by virtual runtime to select the next runnable process, targeting equal CPU share among active tasks. The scheduler operates on the ready queue: processes that have not yet arrived, or that are blocked on I/O, are invisible to the scheduling decision. A task scheduling policy that minimises mean turnaround time must account for the entire job mix, including processes not yet runnable — a horizon that CFS does not have by design.

Separately, CFS is a fairness scheduler, not a throughput scheduler. It makes no attempt to prefer shorter jobs, which is the key property of SRPT, the preemptive optimal policy for minimising mean turnaround time when burst lengths are known. This work assumes full observability of remaining burst — a stronger assumption than any real kernel can satisfy without estimation — in exchange for removing all implicit supervision and studying what a reward signal alone can teach.

2.2 RL for Scheduling: Related Work

KernelOracle uses offline reinforcement learning on kernel scheduling traces to learn a process selection policy, bootstrapped with expert demonstrations from CFS itself. ALPS combines imitation learning with online RL, starting from a behavioural clone of CFS and refining toward a latency objective. Both works encode handcrafted state features — nice values, sleep/wake history, I/O ratios, priority classes — derived from domain knowledge accumulated over decades of scheduler engineering.

Our work differs in three ways: no demonstrations, no handcrafted features, and no target policy. The state is a raw vector of remaining burst, arrival flag, and wait time — features that a real kernel would need to estimate from hardware counters but that require no scheduler-specific domain knowledge to define. We accept stronger assumptions on observability in exchange for removing all implicit supervision from the training signal.

2.3 Related Work: RL Scheduling in Practice

Four deployed or near-deployed systems illustrate the current state of RL-based scheduling and the role that domain knowledge plays in each.

RLScheduler (Zhang et al. 2019) applied deep RL to HPC batch job scheduling, learning a job selection policy from scratch that outperformed hand-tuned priority heuristics on production cluster traces. The critical enabler was a rich per-job state: job type, estimated runtime from historical submissions, queue position, and resource class. The learned policy exploited correlations between job type and actual runtime that no hand-tuned rule had captured. The key difference from this work is observational: RLScheduler had access to historical runtime statistics accumulated over thousands of prior submissions for each job class. Remaining burst was not observed directly — it was inferred from job-type priors.

SCHED (Seo et al. 2025) is the first RL-based CPU scheduling framework designed for direct OS integration. It trains an RL policy offline, then distils the learned policy into a lightweight decision tree for deployment in kernel space, where inference latency budgets are measured in microseconds. The key contribution is a carefully bounded observation space: SCHED restricts state to features computable within the kernel's scheduling tick — sleep/wake history, recent CPU utilisation, voluntary versus involuntary preemption counts — and explicitly excludes features requiring kernel modification to track. The architecture constraint came first; the observation space was designed to fit it.

Double DQN for I/O-Intensive Scheduling (Sun et al. 2025) achieved strong performance on mixed CPU-I/O workloads by extending the state vector to include I/O wait ratio — the fraction of recent time a process spent blocked on I/O — alongside CPU burst estimates. On I/O-intensive task mixes, where CPU burst alone is a poor predictor of actual resource consumption, the I/O wait signal was the decisive feature. Agents trained without it converged to CPU-burst-optimal policies that were suboptimal under the full latency objective.

SmartOS (2021) applied RL to OS resource management across CPU scheduling, memory allocation, I/O prioritisation, and network bandwidth simultaneously, framing the joint allocation problem as a multi-dimensional continuous action space with a single system-level latency reward. The cross-resource coordination — CPU scheduler decisions conditioned on memory pressure and I/O queue depth — produced improvements that no single-resource policy could achieve, at the cost of a substantially more complex observation space encoding interactions across subsystems.

The common thread across successful systems is domain-informed state design: each incorporates features — job type, I/O wait ratio, historical CPU usage, memory pressure — derived from scheduler engineering knowledge accumulated over decades. This work takes the opposite position deliberately: state is restricted to remaining burst, arrival flag, and wait time — features requiring no domain knowledge — to isolate the question of what reward signal alone can teach. The cost of this constraint is visible in the real-trace results (Section 5.15): the agent approaches but does not beat Round Robin on filtered Alibaba trace data. Section 6 discusses the specific features the literature identifies as load-bearing and what adding them would require.

2.4 SRPT as Oracle

Shortest Remaining Processing Time is the preemptive optimal policy for minimising mean turnaround time when burst lengths are known exactly. In our environment, burst lengths are given — the oracle is achievable. On 500 random process sets (master seed 42), SRPT achieves mean MCT = 67.66ms, std = 23.96ms. All random-evaluation performance is reported as deviation from this oracle, not from Round Robin. Round Robin serves only as the lower bar: an agent that fails to beat it has learned nothing useful.

2.5 Queue-Global Information in Production Systems

A 2026 public preview of task queue priority management for cloud batch schedulers introduced a fairness mechanism requiring server-side backlog tracking: the scheduler maintains a global count of queued tasks per job class to detect accumulating delays before they produce starvation. The mechanism explicitly cannot be implemented using per-task metadata alone — the backlog count is not a property of any individual task, but of the queue as a whole. The theoretical framework in Section 3 formalises this observation: we show that any scoring function operating on per-task features is provably unable to make optimal quantum selections for value-preserving schedulers, because both natural scoring objectives (rate-normalised and total value preserved) are monotone functions of quantum size. Breaking this degeneracy requires queue-global information — precisely the server-side backlog the 2026 system tracks. Our empirical ablation (Section 5.23) confirms the theory on synthetic trace data.

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3. Theoretical Framework

3.1 On the Necessity of Queue-Global Information for Value-Aware Scheduling

Value-curve scheduling assigns each task i a value V_i(d) = base_i · max(floor_i, exp(−d/τ_i)), where d is the elapsed delay and τ_i > 0 is the task's urgency parameter. Steep tasks (small τ) lose value quickly; smooth tasks (large τ) decay slowly. A greedy scheduler attempts to select, at each decision step, the action (process, quantum) that maximises value preserved — the gap V_i(0) − V_i(d_i + q), where d_i is the current wait time and q is the proposed quantum.

3.1.1 Potential Surface Formulation

Define the scheduling potential Φ(S) = Σ_i V_i(d_i) over the current queue state S, where the sum runs over all waiting tasks. Each scheduling decision (pick task i, assign quantum q) moves the system along the potential surface. The gradient of Φ with respect to task i's delay is:

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∇Φ_i = −∂V_i/∂d = (base_i/τ_i) · exp(−d_i/τ_i)    [non-plateau]
       = 0                                          [plateau, d_i ≥ τ_i ·
ln(base_i/floor_i)]
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A locally optimal scheduler should select the task with the largest |∇Φ_i| — the task whose value is decaying fastest. This formulation suggests a natural greedy ordering: rank tasks by their current value gradient and serve the steepest.

3.1.2 Process Ordering Impossibility

The gradient ordering requires computing $|\nabla\Phi_i|$ for every queued task simultaneously and comparing them. This is an inherently queue-global operation: the selection of task i depends not on i's features alone, but on whether any co-queued task j has $|\nabla\Phi_j| > |\nabla\Phi_i|$. Any policy that assigns a score $f_{order}(base_i, floor_i, \tau_i, d_i)$ to task i in isolation and selects $\arg\max_i f_{order}$ cannot, in general, produce the correct ordering — because the same score for task i may be optimal in one queue composition and suboptimal in another, depending on the co-queued population.

On traces dominated by tasks with similar τ (as in the Alibaba batch_task data, where 92% of tasks have $\tau_{smooth} \approx 800s$), the gradients $\nabla\Phi_i$ are nearly identical across the queue: the ratio $\nabla\Phi_i / \nabla\Phi_j = \exp(-(d_i - d_j)/\tau) \to 1$ as $\tau \to \infty$ relative to delay differences. In the empirical evaluation (Section 5.22), 93–100% of ordering decisions are near-ties (top-2 gradients within 10%) regardless of τ calibration. The potential surface is flat over the vast majority of the trace composition, making gradient-based ordering informationally void on this data.

3.1.3 Quantum Selection Impossibility

Even if process ordering were resolved by queue-global comparison, quantum selection presents a separate impossibility. Given a fixed task i with delay d_i, a scheduler must choose quantum q from a discrete set $\{q_{min}, q_{med}, q_{max}\}$. Two natural per-task scoring objectives are:

$f_{rate}(q) = (V_i(d_i) - V_i(d_i + q)) / q$ — value preserved per unit time.

$f_{total}(q) = V_i(d_i) - V_i(d_i + q)$ — total value preserved by the action.

Lemma (Quantum Scoring Monotonicity). For any $\tau > 0$, $base > 0$, $floor \in [0, 1)$, and $d \geq 0$: (a) $f_{rate}(q)$ is strictly decreasing in q for all $q > 0$. (b) $f_{total}(q)$ is strictly increasing in q for all $q > 0$ such that $d + q$ is below the plateau.

Proof. (a) Let $h(q) = (1 - e^{-q/\tau})/q$. Then $f_{rate}(q) = (base/\tau) \cdot h(q) \cdot e^{-d/\tau} \cdot \tau/1 = base \cdot e^{-d/\tau} \cdot h(q)$. It suffices to show h(q) is strictly decreasing. Differentiating: $h'(q) = [(q/\tau + 1)e^{-q/\tau} - 1] / q^2$. Setting $u = q/\tau > 0$, the numerator is $(u + 1)e^{-u} - 1$. Since $(u + 1)e^{-u} < 1$ for all $u > 0$ (as $e^u > 1 + u$ by strict convexity), the numerator is strictly negative. Therefore $h'(q) < 0$ and f_{rate} is strictly decreasing. □

b. $f_{total}(q) = base \cdot e^{-d/\tau} \cdot (1 - e^{-q/\tau})$. Since $1 - e^{-q/\tau}$ is strictly increasing in q for $\tau > 0$, f_{total} is strictly increasing. □

Corollary (Queue-Global Necessity). No per-task scoring function — operating solely on $(base_i, floor_i, \tau_i, d_i, q)$ — can produce mixed quantum selection across the discrete set $\{q_{min}, q_{med}, q_{max}\}$. f_{rate} always selects q_{min}; f_{total} always selects q_{max}. Any other scoring function that selects an intermediate quantum must encode information about the queue state beyond task i's features alone. Queue-global information is therefore necessary for optimal quantum selection in value-aware scheduling.

3.1.4 Queue-Global Resolution

The impossibility results motivate a scheduling architecture that maintains queue-global state: at each decision step, the scheduler maintains a running estimate of the queue composition (e.g., the distribution of τ values and current delays) and uses this estimate to inform both ordering and quantum selection. The two-head attention mechanism introduced in Week 10C (Section 5.20) is one instantiation of this principle: one attention head specialises in competitor context (queue state) and the other in selection (per-task scoring). The theoretical framework explains why single-head attention is insufficient — a single head that produces both context aggregation and selection scoring collapses to a per-task proxy under the monotonicity constraint.

3.1.5 Empirical Confirmation

The theoretical predictions are confirmed by three ablations reported in Section 5.23: (1) a quantum_only policy using f_rate selects q_min 100% of the time across both default and calibrated τ initializations; (2) a quantum_only_fixed policy using f_total selects q_max 100% of the time; (3) an ordering_only policy using gradient ordering with fixed q_med achieves total_value = 15.37, representing near-complete starvation of smooth tasks — confirming that per-task ordering, even when correctly implemented, cannot resolve the value preservation objective without queue-global quantum selection. The full value_aware policy (total_value = 250.55) requires both ordering and quantum selection informed by the queue state to avoid degeneracy.

4. Method

4.1 Environment

Five processes per episode. Arrival times are sampled without replacement from {0, 2, 5, 8, 10}ms; burst lengths from U[1, 60]ms, re-sampled each episode (randomised from Week 2 onward; fixed in Week 1). The action space is 15 actions: 5 processes × 3 quantum tiers (1ms, 5ms, 20ms). Invalid actions — processes not yet arrived, or completed — are masked at every decision step. An episode ends when all processes complete.

Baselines. Round Robin (5ms fixed quantum, PID-ascending cycle): MCT ≈ 34.4ms (in-distribution), 36.4ms (OOD). SRPT oracle: MCT ≈ 26.4ms (in-distribution), 28.4ms (OOD).

4.2 Reward Derivation

The reward signal at each decision step is:

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R(t) = −n_active(t) × q_actual(t)
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where n_active(t) is the number of processes that have arrived and not yet completed, and q_actual = min(quantum, remaining_burst) is the CPU time actually consumed.

By Little's Law, mean turnaround time equals the time-average number of active processes times the mean time a process spends in the system. Minimising the cumulative sum of n_active × q_actual — the area under the active-count curve — is therefore equivalent to minimising total turnaround time. The reward requires no engineering of completion bonuses, shaped incentives, or domain-specific terms. It is theoretically principled from a queuing standpoint.

From Week 3, the reward is normalised by REWARD_SCALE = 100.0. The maximum raw magnitude per step is N × max_quantum = 5 × 20 = 100, so normalisation maps each step to [−1, 0], keeping cumulative Q-value targets in approximately [−150, 0] — a range representable without numerical divergence. An earlier training run without normalisation confirmed this need: loss grew from 46 to 87,799 over 10,000 episodes, Q-values reached +3,000.

4.3 State Representation

Weeks 1–2 (tabular). State is a 5-tuple of discretised remaining-burst bins. Each process is mapped to one of 7 bins: 0 (complete), 1–5 (active tiers defined by burst-length thresholds), 6 (not arrived). The Q-table has shape (7,7,7,7,7,15) ≈ 1.9MB, supporting 16,807 distinct states.

Weeks 3–4 (DQN). State is a 15-dim continuous float32 vector — per process: [remaining_burst/60, arrived_flag, wait_time/300]. This encodes full burst magnitude, eliminating the discretisation ceiling.

Week 5 (action-conditioned). The 15-dim state is retained but restructured into a 19-dim network input: the 15-dim state with the candidate's PID slot zeroed (s_masked, 15-dim), concatenated with the candidate's own features plus quantum tier (a4 = [remaining/60, arrived_flag, wait/300, qt/2.0], 4-dim). The zeroing step removes redundancy and is what makes the permutation invariance argument exact (see Section 4.5).

4.4 Architecture Evolution

Tabular (Weeks 1–2). Q-table indexed by 5-tuple of discretised bins. $\alpha = 0.1$, $\gamma = 1.0$, ε decaying from 1.0 to 0.05 over 10,000 episodes via linear schedule.

DQN (Weeks 3–4). 15 → 64 (ReLU) → 32 (ReLU) → 15 (linear), 3,599 parameters. He initialisation. Adam (lr=0.001, β_1=0.9, β_2=0.999). Global gradient norm clipping (max_norm=1.0). Experience replay buffer (capacity 10,000, batch 32). Target network hard-copied every 200 episodes. 500-transition warmup of random exploration before first gradient update.

Action-conditioned DQN (Week 5). 19 → 64 (ReLU) → 32 (ReLU) → 1 (linear), 3,393 parameters. Same optimiser, same replay and target schedule. At each action selection step, 15 separate (s_masked ‖ a4) vectors are constructed — one per candidate action — and evaluated in a single batched forward pass. At training time, Q-values for all 15 next-state actions are computed in one (batch × 15, 19) = (480, 19) matrix multiply against the target network, keeping wall-clock training time within approximately 1.5× of Week 3 despite the 15× query expansion.

4.5 The Masking Amendment and Its Rationale

An initial Week 5 design included the candidate's own features in both s_masked and a4, creating redundant information at two positions in the input vector. The amendment zeroes the candidate's PID slot in s_masked before concatenation.

This is not a tidying change — it is what makes the permutation invariance argument exact. With the candidate's slot zeroed, the s_masked vector for (P0 as candidate, competitors at P1–P4) and (P4 as candidate, competitors at P0–P3) are identical whenever the competitor burst magnitudes are identical and the competitors occupy the same PID positions. The a4 vector is identical when candidate burst, arrival status, and quantum tier are identical. Identical inputs produce identical outputs. Without the zeroing, the candidate's features appear at two positions in the 19-dim input vector — once in s_masked at its PID index and once in a4 — so P0 and P4 inputs can never be identical even when all process features match.

The formal invariance guarantee: for any pair of candidate PIDs (p, p') and any state where both have the same remaining burst, arrival flag, and wait time, and where all competitors are at the same PID positions with the same features, Q(s, p, qt) = Q(s, p', qt) exactly.

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5. Results

5.1 Twelve-Week Performance Summary

Random evaluation — 500 process sets, master seed 42, same seeds for all agents:

Agent	Mean MCT	Std	vs SRPT	Eval set
SRPT oracle	67.66ms	23.96ms	0.00ms	Random (500)
DQN W5 (action-conditioned)	76.04ms	27.70ms	+8.38ms	Random (500)
DQN W7 (attention-annealed)	80.57ms	21.82ms	+12.91ms	Random (500)

Agent	Mean MCT	Std	vs SRPT	Eval set
DQN W3 (position-indexed)	85.00ms	26.51ms	+17.34ms	Random (500)

Fixed-set evaluation — two specific process configurations from Week 1 (not seen during W3/W5 training):

Agent	Mean MCT	Std	vs SRPT	Eval set
DQN W5 (in-dist fixed)	31.00ms	0.00ms	+4.60ms	Fixed in-dist
DQN W5 (OOD fixed)	29.60ms	0.00ms	+1.20ms	Fixed OOD
DQN W3 (in-dist fixed)	34.60ms	0.00ms	+8.20ms	Fixed in-dist
DQN W3 (OOD fixed)	32.00ms	0.00ms	+3.60ms	Fixed OOD
Tabular W2 (in-dist)	29.80ms	0.00ms	+3.40ms	Fixed in-dist
Round Robin	34.40ms	—	+8.00ms	Fixed in-dist

Filtered real-trace evaluation — 500 filtered Alibaba 2018 episodes, master seed 42 (Weeks 8–12):

Agent	Mean MCT	Std	vs Round Robin	Primary metric
SRPT oracle (filtered)	15.20s	8.94s	—	MCT
W10C 2-head attention	18.21s	12.15s	**−3.10s**	MCT
W9 1-head fixed β	19.22s	11.58s	−2.09s	MCT
W10C-VC2 ep 5000 (value-curve)	21.46s	—	+0.15s	total_value
Round Robin (1.0s, filtered)	21.31s	14.03s	baseline	—
W10C-VC1 ep 2000 (value-curve)	26.56s	—	+5.25s	total_value

The three blocks use different evaluation protocols and must not be compared row-to-row. The tabular agent's MCT of 29.80ms on the in-distribution fixed set reflects effective memorisation of that configuration across 10,000 training episodes; it cannot participate in random evaluation because std = 0.00ms reveals discretisation collapse rather than a generalisable policy. The VC1 and VC2 agents optimise total value preserved rather than MCT; their MCT figures are secondary metrics reported for comparability, not primary optimisation targets.

5.2 F1 — Zero Variance Is Not Quality

The Week 1 tabular agent achieves MCT = 29.80ms on its training set — within 3.40ms of SRPT — and beats Round Robin by 13.5%. By fixed-set metrics, this looks like successful learning.

The failure is revealed by std = 0.00ms on both evaluation sets. Two process sets that differ only in their within-bin burst magnitudes — say, one process at 3ms and another at 8ms, both landing in bin 1 — receive identical state tuples and therefore identical Q-values and identical action sequences. The agent cannot distinguish them. Any reported variance would require burst differences to cross a bin boundary.

Identity bias confirms the structural diagnosis. In the probe state (P1 = 10ms remaining, P4 = 3ms remaining, all others complete or not arrived), SRPT prefers P4. The Week 1 tabular agent assigns Q(P1/long) > Q(P4/long), |Δ| = 0.270, wrong direction. Week 2 randomised training corrects the direction (|Δ| = 0.225, correct) but cannot shrink the gap further — the Q-table is indexed by PID, so P1 and P4 always occupy different weight vectors regardless of how similar their burst magnitudes are. The bias is not a training problem; it is a representation problem.

Conclusion: std = 0.00ms is a sign of discretisation collapse, not convergence. Fixed-set evaluation cannot detect this; random-set evaluation with diverse burst magnitudes is the minimum diagnostic.

5.3 F2 — DQN Unlocks Variance, Introduces New Failure Mode

The Week 3 DQN produces non-zero variance immediately: std = 26.51ms across 500 random episodes, compared to SRPT's std = 23.96ms. The difference of 2.55ms represents policy noise on top of the structural (process-set) variance that both agents inherit. The network is adapting to each episode, not replaying a memorised schedule.

But the mean is 85.00ms against SRPT's 67.66ms — a gap of 17.34ms. And identity bias worsens substantially:

Agent	Q(P1/long)	Q(P4/long)	Δ	
Tabular W1	—	—	0.270	No
Tabular W2	−3.91	−3.68	0.225	Yes
DQN W3	+44.92	+39.43	5.48	No

The move to continuous state did not reduce positional bias — it amplified it. The 15 output neurons are permanently assigned to PID positions and accumulate asymmetric gradient histories across 10,000 episodes. Gradient updates for (P0 = 10ms, P4 = 3ms) and (P2 = 10ms, P1 = 3ms) flow through completely disjoint weight paths. The network has no mechanism to learn "prefer the shorter process" as a general rule; it can only learn "when P0 has 10ms and P4 has 3ms, prefer P4" — a rule that does not transfer.

The loss curve also rises between episodes 5,000 and 10,000 (0.0398 → 0.6504), suggesting the 15 simultaneous regression targets create a noisy multi-task learning problem with cross- contaminated gradient paths.

5.4 F3 — Policy Analysis Diagnoses the Artefact

Week 4 extracts 10,000 greedy decisions from the trained Week 3 DQN and compares each to the SRPT oracle, grouped by n_active. The results reveal two patterns.

First, process-selection agreement is load-dependent: 100% at n_active = 1 (trivially correct — only one valid choice), dropping to the high-50% to low-60% range as more processes compete. Better than random (20%), well below SRPT.

Second, and more striking: the quantum selection pattern shifts with load. At n_active = 1 the network uses predominantly 1ms quanta, matching SRPT. But as n_active grows, 20ms quanta dominate, used in more than 60% of decisions at n_active ≥ 3. This pattern is coherent, reproducible across evaluation runs, and superficially plausible — large quanta amortise context-switch overhead when many processes are waiting.

The interpretation is wrong. The pattern is a PID-positional artefact. Specific PID combinations that co-occur frequently at high n_active values during training accumulate gradient pressure toward large quanta in those position-indexed action heads. This pressure is absent in other PID orderings. The pattern looks load-dependent because load and PID co-occurrence are correlated in the training distribution, not because the network learned load-responsive behaviour. The Week 5 architecture test, which shares weights across all PID positions, eliminates the pattern entirely.

The methodological point is general: for any RL agent with position-indexed action heads, policy patterns that correlate with action position should be treated as artefact candidates until a weight-sharing ablation rules out positional origin.

5.5 F4 — Action-Conditioned Architecture Resolves Bias and Eliminates Artefact

Loss curve. The Week 5 loss converges to 0.0001 by episode 10,000, compared to 0.6504 for Week 3. A single scalar regression target per action, rather than 15 simultaneous targets with shared gradient paths, produces a more coherent learning problem.

Episode	Week 3 Avg Loss	Week 5 Avg Loss
500	0.0219	0.0114
1,000	0.0082	0.0026
2,000	0.0067	0.0007
5,000	0.0398	0.0002
10,000	0.6504	0.0001

Random evaluation. Mean MCT = 76.04ms (std = 27.70ms; p5 = 34.17ms; p95 = 121.51ms; min = 17.39ms; max = 166.99ms). The 8.96ms improvement over Week 3 (85.00ms) — a 10.5% reduction — comes from the architecture change alone: same environment, same reward, same training budget, same hyperparameters.

Identity bias. The probe is re-run with the Week 5 architecture:

Agent	Q(P1/long)	Q(P4/long)	Δ	
Tabular W1	—	—	0.270	No
Tabular W2	−3.91	−3.68	0.225	Yes
DQN W3	+44.92	+39.43	5.48	No
DQN W5	−0.825	−0.774	0.051	Yes

$|\Delta| = 0.051$, correct direction. The smallest gap across all four agents, despite the residual being expected: the competitor (P1) is at a fixed PID position in the probe state, and the partial invariance guarantee applies to the candidate, not the competitor context.

Permutation invariance unit test. Two probe states constructed with identical competitor features but different candidate PIDs (P0 vs P4): - Q(P0 as candidate, P1 as sole competitor at 10ms, qt=1ms) = **−0.9054024509** - Q(P4 as candidate, P1 as sole competitor at 10ms, qt=1ms) = **−0.9054024509** Exact equality confirmed. The masking amendment produces provably identical inputs and therefore identical outputs for any two candidates in structurally equivalent positions.

Policy analysis (Week 5). The 20ms quantum preference is gone.

n_active	agree%	1ms%	5ms%	20ms%
1	100.0%	90.0%	7.6%	2.4%
2	74.3%	99.3%	0.0%	0.7%
3	65.4%	99.9%	0.1%	0.0%
4	67.9%	99.9%	0.0%	0.1%
5	71.9%	99.9%	0.0%	0.1%

At n_active ≥ 2, the agent uses 1ms quanta in >99% of decisions — matching the SRPT oracle's quantum strategy exactly. The Week 3 load-adaptive quantum pattern was not a heuristic; it was PID-positional noise that disappeared the moment weights were shared across positions.

Process-selection agreement improved at n_active = 2 (from ~65% in Week 3 to 74.3% in Week 5) but did not reach the 80% target. Agreement plateaus at 65–68% for n_active ≥ 3, and the loss curve shows convergence, ruling out a training-time explanation.

5.6 F5 — Attention Collapses Without Entropy Regularisation

Week 7 introduces AttentionDQN: a dot-product attention head over competitor encodings replaces the flat s_masked concatenation used in Week 5. The candidate query vector q = cand_enc @ W_Q + b_Q attends over competitor key-value pairs (K = comp_encs @ W_K + b_K, V = comp_encs @ W_V + b_V), producing a context vector that is a weighted sum of competitor value encodings. The MLP then receives [context ‖ a4] — a 12-dim input — rather than [s_masked ‖ a4] (19-dim). Full permutation invariance over the competitor set follows from the softmax operation: any permutation of competitor inputs produces the same weighted sum and therefore identical outputs. Permutation invariance unit test: $|\Delta| = 0.000$ (W7) vs $|\Delta| = 0.0335$ (W5).

Without regularisation (λ=0), attention entropy H collapses from 1.386 (uniform over 4 competitors) to H = 0.03 by episode 1,000. The softmax locks onto a single competitor early in training — before the TD signal has accumulated enough gradient to teach which competitor to focus on. Once near-1-hot, the attention Jacobian approaches zero, and the weight parameters stop receiving meaningful gradient. The result is a hardcoded-focus policy that ignores three of four competitors: mean MCT = 117.51ms, SRPT agreement = 21.9%, below the Round Robin lower bar.

Two fixed-λ runs confirmed the failure is not trivially solvable. λ=0.05 stabilised H at 0.66–0.67 but the entropy term dominated the TD signal, producing negative total loss, agreement = 64.6%, mean MCT = 87.73ms. λ=0.10 worsened the domination: H stable 0.66–0.71, agreement fell to 57.4%, mean MCT rose to 94.41ms. The entropy regulariser was functioning as the primary learning signal, not the auxiliary stabiliser it was intended to be.

Entropy annealing resolved the conflict: $\lambda(ep) = 0.10 - (0.10 - 0.005) \times (ep / 10000)$. The regulariser prevents collapse during early episodes while shrinking to near-zero once the TD signal is stable. H held in the range 0.60–0.68 throughout 10,000 episodes; total loss converged near zero. The annealed run is the Week 7 trained agent.

Finding: Attention entropy collapse is a failure mode distinct from both discretisation collapse (Week 1) and position-indexed gradient contamination (Week 3). It requires active regularisation with an annealed schedule: a fixed λ that prevents collapse also dominates the TD signal; only a decaying λ threads the needle between softmax freeze and TD-signal erasure.

5.7 F6 — Performance Regression in Mean, Improvement in Variance

The annealed AttentionDQN achieves mean MCT = 80.57ms, std = 21.82ms on 500 random episodes. Comparing the three DQN generations alongside the oracle:

Agent	Mean MCT	Std	vs SRPT
SRPT oracle	67.66ms	23.96ms	0.00ms
DQN W5 (action-conditioned)	76.04ms	27.70ms	+8.38ms
DQN W7 (attention-annealed)	80.57ms	21.82ms	+12.91ms
DQN W3 (position-indexed)	85.00ms	26.51ms	+17.34ms

Mean MCT regressed from W5 (76.04ms) to W7 (80.57ms) by +4.53ms. But std fell from 27.70ms to 21.82ms — below the SRPT oracle's own std of 23.96ms. W7 is more consistent than SRPT while being 12.91ms slower in mean. This is not a net regression; it is a quality tradeoff between two distinct policy properties. W5's higher mean agreement with SRPT includes tail decisions that reach large errors (p95 = 121.51ms); W7's attention context suppresses those tail errors — at the cost of some mean throughput. A policy with std < SRPT while maintaining a positive SRPT gap implies the agent is trading median throughput for reduced worst-case exposure.

Identity bias is preserved in the correct direction: Q(P1/long) = −2.434, Q(P4/long) = −2.413, $|\Delta| = 0.021$, correct direction. The residual gap is smaller than W5's $|\Delta| = 0.051$, consistent with full permutation invariance over the competitor context (the probe state's single competitor is now aggregated without positional weighting).

Finding: Mean MCT regression accompanied by std < SRPT oracle represents a changed quality tradeoff, not a net regression. The appropriate evaluation metric for comparing W5 and W7 is the joint distribution of (mean, std), or equivalently the p95 tail performance — not mean alone.

5.8 F7 — Process Selection Meets 80% Target at n_active ≤ 3, Fails at n_active = 5

Policy analysis on 10,000 greedy decisions from the annealed W7 agent, compared to W5:

n_active	W7 agree%	W5 agree%
1	100.0%	100.0%
2	85.8%	74.3%
3	90.9%	65.4%
4	71.7%	67.9%
5	40.7%	71.9%

The 80% process-selection target — not met by W5 at any n_active ≥ 2 — is met and exceeded by W7 at n_active = 2 (85.8%) and n_active = 3 (90.9%). But agreement falls sharply to 71.7% at n_active = 4 and collapses to 40.7% at n_active = 5 — worse than W5's 71.9% on the same load condition.

The n_active = 5 failure has a direct architectural explanation. With 4 active competitors and attention entropy $H \approx 0.62$ at episode 10,000, each competitor receives approximately 0.22–0.25 weight — near-uniform. When attention weights are near-uniform, the context vector approaches mean-pooling over competitor encodings. Mean-pooling loses relative ordering information: it cannot distinguish "three competitors at 30ms plus one at 5ms" from "four competitors at 20ms each." At n_active = 5, the attention mechanism degenerates toward the sum-pooling baseline from which AttentionDQN was intended to advance.

The residual entropy $H \approx 0.62$ is the proximate cause: the annealed λ prevented collapse but did not drive the agent to sufficiently sharp attention under high competitor count. The final $\lambda = 0.005$ is not small enough to allow further softmax sharpening at n_active = 5. This is a learnable-temperature problem, not a training-duration problem. Longer training with the same fixed-temperature softmax would maintain the same H floor; what is needed is a parameter that can sharpen the distribution independently of the entropy annealing schedule.

Finding: The 80% SRPT-agreement target is met at n_active ≤ 3 but not at n_active ≥ 4. The n_active = 5 failure is attributable to residual entropy at termination: $H \approx 0.62$ produces near-mean-pooling at 4 competitors, erasing the ordering information the attention was introduced to capture. The fix is a learnable inverse-temperature parameter β (replacing fixed $1/\sqrt{d}$ scaling with $1/\sqrt{d} \times \exp(\beta)$, where β is trained alongside the attention weights) that can sharpen independently of the entropy annealing schedule.

5.9 F8 — Attention Tracks "Threatening Competitor," Not Selected Process

The initial attention diagnostic produced an apparent contradiction: in 80.2% of decisions the highest attention weight lands on the longest competitor, while in 82.2% of decisions the DQN agrees with SRPT — which always selects the shortest process. If attention were guiding selection toward the longest competitor, agreement with SRPT would be low, not high.

A joint diagnostic on 1,000 decisions (n_active ≥ 3) resolves the contradiction. At each decision, the DQN-selected process, the SRPT-selected process, and the highest-attention competitor are recorded jointly:

Case	n	Attention on shortest (%)	Attention on longest (%)
DQN agrees with SRPT	660	4.1%	90.6%
DQN disagrees with SRPT	340	12.4%	62.4%

When the DQN correctly selects the shortest process, every competitor is by definition longer. "Highest attention on longest competitor" and "highest attention on most distant competitor" are the same statement. The 90.6% longest-focus in agree cases means the attention is locking onto the structurally extreme competitor — the one whose remaining burst anchors the upper end of the competitive context — while the MLP uses that context to confirm the correct short-process selection.

The disagree cases confirm the asymmetry is diagnostic: 62.4% longest-focus when the agent errors, versus 90.6% when correct. Sharper attention to the most threatening competitor corresponds to better decisions. The attention is not misfiring toward a long process instead of the correct short one; it is providing maximum-contrast context that the MLP interprets correctly.

Sanity check: 660/1,000 = 66.0% SRPT agreement in the joint diagnostic vs 82.2% in full policy analysis. The difference reflects the n_active ≥ 3 filter — the joint diagnostic excludes n_active = 1 (trivially 100%) and n_active = 2 (85.8%), leaving the harder subset.

Finding: Attention-on-longest in correct decisions is not a misalignment with SRPT. The attention learned to track the most threatening competitor (longest remaining burst) as structural context for a correct short-process selection. The apparent contradiction between longest-focus and SRPT agreement dissolves once conditioning on decision correctness is applied: the pattern is stronger in correct decisions (90.6%) than in errors (62.4%), indicating attention focus quality correlates with decision quality.

5.10 F9 — Distribution Mismatch Is the Root Cause of Trace Transfer Failure

The synthetic training distribution (U[1, 60]ms) has a p95/p50 ratio of approximately 1.9×. The Alibaba 2018 batch_task trace has p95/p50 = 39.7×. These are not the same problem at different scales. The p50 of the trace is 10s; the p95 is 397s; the max is 583,886s — a span of four orders of magnitude. Under linear normalization (`burst / BURST_P95`), the median task maps to 10/397 = 0.025 while 27.7% of sampled 5-task episodes contain at least one task with normalized burst > 0.90, near the clipping ceiling. The remaining 73% of tasks cluster below 0.12 in normalized feature space. Key-query dot products for the outlier task are an order of magnitude larger than those for the short tasks, saturating the softmax and destroying ordinal information among the non-outlier tasks in every episode that contains an extreme value.

This failure is not architectural. The AttentionDQN's permutation invariance guarantee, entropy annealing mechanism, and key-query-value structure are all intact. The failure is a training distribution failure: a normalization scheme effective for 1.9× p95/p50 is insufficient for 39.7×. A clipping point that represents the distributional center for a near-uniform distribution represents only the 95th percentile tail for a power-law distribution — the two cases require qualitatively different normalization strategies.

Finding: The transfer failure from synthetic to real-trace data is a normalization failure, not an architecture failure. Linear normalization by p95 is appropriate when most of the distribution is near the normalization constant; it fails when the normalization constant is a rare extreme that the majority of inputs never approach.

5.11 F10 — The Trace-Trained Agent Failed to Transfer: Results

The W8 agent trained from scratch on full Alibaba trace episodes, with all W7 hyperparameters intact, is evaluated on 500 trace test episodes (master seed 42):

Agent	Mean MCT	Std	vs RR	vs SRPT
W8 AttentionDQN (linear norm)	135.76s	274.99s	+26.24s	+45.63s
SRPT oracle (trace)	90.13s	191.25s	—	0.0s
Round Robin (2.0s, trace)	109.52s	202.78s	base	—

The agent is 26.24s worse than Round Robin in mean MCT. Policy analysis on 10,432 greedy decisions: overall SRPT agreement 50.2%, collapsing to 26–31% at n_active ≥ 2 — near the random baseline of 25% for a 4-

candidate selection. Identity bias probe: $|\Delta| = 0.051$, wrong direction (agent prefers the longer process in the two-process case). Attention entropy: H = 0.13–0.24 throughout all 10,000 training episodes — collapsed from the first checkpoint.

The W8b variant (log-normalization, REWARD_SCALE = 40.0, λ: 0.30 → 0.005) produced worse results: mean MCT = 157.13s, SRPT agreement 23.4%, H = 0.13–0.20. Log-normalization alone, without addressing the within-episode burst ratio, did not prevent attention collapse.

Finding: Training directly on full-trace episodes without correcting the within-episode magnitude structure produced no learning signal for SRPT process selection. This is not a partial transfer with degraded performance; it is no transfer.

5.12 F11 — Attention Entropy Collapsed on Full Trace Data

The W7 agent trained on synthetic data stabilized at H ≈ 0.62 throughout 10,000 episodes under the annealed regularizer. The W8 agent trained on full-trace data produced H = 0.13–0.24 throughout — below the 0.20 collapse flag from the first checkpoint and remaining there for all 10,000 episodes, including with λ increased from 0.10 to 0.30.

The trace's 39.7× p95/p50 skew creates attention collapse pressure that the annealing schedule cannot overcome. When a 5-task episode contains one task at normalized burst 0.90 and four tasks at 0.025, the key-query dot products for the outlier are approximately 36× larger than for the short tasks. The softmax exponential amplifies this into a near-zero weight for all non-outlier competitors. The entropy regularizer must generate gradient large enough to redistribute weight from a competitor with near-zero Jacobian — a task that would require λ large enough to dominate the TD signal entirely.

Finding: Entropy collapse on heavy-tailed data is not solvable by increasing λ alone. A distribution where outliers dominate key-query dot products by a factor of 36× creates collapse pressure that is qualitatively different from the 1.9× synthetic case. The required resolution is a normalization scheme that equalizes feature magnitudes before the attention computation — not a stronger regularizer fighting a structurally hostile input distribution.

5.13 F12 — What Transfer to Real Traces Requires

Three changes are necessary for real-trace generalization. Each is precisely stated.

1. Log-normalization of burst durations. Replace `burst / BURST_P95` with `log(1 + burst) / log(1 + BURST_P95)`. This maps the full trace range to approximately [0, 1] with the heavy tail compressed. The median task (10s) maps to 0.40 rather than 0.025 under linear normalization. The key-query dot-product ratio between outlier and short tasks falls from ~36× to ~2.5×, within the range the entropy regularizer can handle. This is a one-line change to `_encode_state`.

2. Within-episode magnitude control via trace filtering. Log-normalization is necessary but not sufficient when the full distribution has p95/p50 = 39.7×. Filtering tasks above the 75th percentile duration reduces the within-episode burst ratio from p50 = 74× (random sampling, full trace) to p50 = 14× (random sampling, filtered trace). This is the minimum viable preprocessing step for attention-based RL scheduling on this workload.

3. Recalibrated entropy regularization. The annealing schedule λ: 0.10 → 0.005, calibrated for the 1.9× synthetic distribution, is insufficient for the residual skew of even the filtered trace. The initial λ must increase to at least 0.30, with the same decay schedule, to maintain H > 0.20 during early training. Alternatively, a learnable temperature parameter β (replacing fixed $1/\sqrt{d}$ with trainable $\beta/\sqrt{d}$) removes the manual calibration requirement.

5.14 F13 — Filtering Enables Real-Trace Training

The full Alibaba 2018 batch_task training split (11,436,584 tasks, p95/p50 = 39.7×) produced attention collapse in every training configuration tested. Filtering tasks above the 75th percentile duration (47.0s, computed from the training split) reduced the dataset to 8,076,473 tasks (70.6% retained) and changed the within-episode distribution qualitatively:

Metric	Full trace	Filtered trace
p95/p50 ratio	39.7×	7.2×
Within-episode p50 burst_ratio	74.0×	14.0×
Within-episode % > 10×	89.7%	64.1%
Within-episode % < 3×	0.4%	2.0%

The filtered test split retains 2,151,487 of 2,859,147 rows (75.2%). The new distribution constants from the filtered training split: BURST_P95 = 36.0s, BURST_MEDIAN = 5.0s, quantum tiers (0.25s, 1.0s, 4.0s). The W8c agent trained on filtered episodes with log-normalization and λ: 0.30 → 0.005 stabilized attention entropy at H = 0.32–0.50 throughout 10,000 episodes — compared to H = 0.13–0.24 on the full trace with all three fixes applied. Training time dropped from 56 minutes (full trace, 10M+ tasks per epoch) to 6 minutes (filtered, 8M tasks).

Finding: Filtering tasks above the p75 duration threshold is the minimum viable preprocessing step for attention-based RL scheduling on the Alibaba 2018 trace. It reduces the within-episode p50 burst ratio from 74× to 14× and enables the entropy annealing mechanism to function as designed. It does not produce a representative sample of the full workload — the top 30% of the duration distribution is excluded — but it produces a tractable training distribution for the current architecture.

5.15 F14 — W8c Performance on Filtered Trace

The W8c agent (filtered trace, log-normalization, λ: 0.30 → 0.005, REWARD_SCALE = 20.0) is evaluated on 500 filtered test episodes (master seed 42):

Agent	Mean MCT	Std	vs RR	vs SRPT
SRPT oracle (filtered)	15.72s	8.88s	—	0.0s
W8c AttentionDQN	23.30s	10.85s	+1.53s	+7.58s
Round Robin (1.0s, filtered)	21.77s	13.61s	base	—

Two observations require honest framing.

First, the agent is 1.53s worse than Round Robin on mean MCT. The gap is well within one standard deviation of Round Robin's std (13.61s), and the agent's std (10.85s) is lower than Round Robin's — meaning the agent's distributional tail is better controlled even when its mean is marginally higher. This is the same quality tradeoff identified in W7 (mean MCT regression accompanied by std improvement). It is not a resounding success; the agent has not learned to beat the simplest non-trivial baseline on real data.

Second, process-selection agreement collapses at n_active ≥ 3: 100% at n_active = 1 (trivial), 55.3% at n_active = 2, 47.1% at n_active = 3, 21.3% at n_active = 4, 17.3% at n_active = 5. This is a pattern similar to W7's attention collapse — agreement falls sharply with competitor count — but here H = 0.32–0.50 throughout training. The attention did not collapse to noise; it learned a coherent but wrong strategy. The identity bias probe confirms the wrong direction: Q(candidate/tier2) = −3.132, Q(competitor/tier2) = −2.976, |Δ| = 0.156, candidate not preferred. Overall SRPT agreement: 57.9% across 10,035 decisions.

Finding: The W8c agent achieves mean MCT within 1.53s of Round Robin with lower variance — a marginal performance result that represents a 94% reduction in the performance gap compared to the unfiltered W8a baseline (+26.24s vs Round Robin). The filtering, log-normalization, and entropy recalibration are necessary conditions for tractable training. They are not sufficient for competitive performance: process-selection accuracy at n_active ≥ 3 remains near-random, not from entropy collapse but from a learned anti-SRPT attention strategy.

5.16 F15 — Attention Attends to Longest Competitor on Real Traces

The W8c attention diagnostic, applied to 1,006 decisions at n_active ≥ 3:

Metric	W8c (filtered trace)	W7 (synthetic) — agree cases
Highest attention = longest competitor	93.7%	90.6%
Highest attention = shortest competitor	4.1%	4.1%
Mean weight on longest	0.5132	—
Mean weight on shortest	0.3231	—

The pattern is superficially identical to W7's attention-on-longest finding (Section 5.9). But the interpretation differs in a critical way. In W7, attending to the longest competitor was structurally correct: when the agent selects the shortest process, all competitors are by definition longer, so 90.6% longest-focus in agree cases is a tautological consequence of correct selection. The pattern correlated with quality (90.6% in agree cases vs 62.4% in disagree cases).

In W8c, the agent is agreeing with SRPT in only 57.9% of decisions and its identity bias probe points in the wrong direction. The 93.7% longest-competitor attention here is not a marker of correct short-process selection — it is the agent's primary decision signal, attending to the longest competitor to schedule it next or to avoid it in a way that generates the observed agreement with chance. The mean weight on the longest competitor (0.5132) versus the shortest (0.3231) shows the attention is not near-uniform; it is deliberately concentrating on the longest task.

The cause is the area-integral reward signal. The reward `−n_active × q_actual` penalizes every unit of CPU time consumed while any process is waiting. Under this signal, the locally optimal strategy when facing multiple competitors is to attend to the one with the most remaining burst — the one that will contribute the longest waiting penalty to future steps — and either schedule it first to eliminate its influence, or use its magnitude as a reference point for comparison. The reward creates gradient pressure toward longest-competitor attention that is locally coherent under the objective but anti-SRPT globally: SRPT requires scheduling the *shortest* process to minimize mean completion time, which reduces n_active fastest, but the reward signal does not differentiate between "run the short process now, reducing n_active" and "run the long process now, using its magnitude as context." Both reduce n_active by 1 eventually; only SRPT minimises the area integral over all processes jointly.

The W7 synthetic agent avoided this failure not because of a different reward signal but because the synthetic uniform distribution produced a different gradient landscape — the attention learned to use longest-competitor context as a reference for *contrast* rather than as a selection target. That distinction did not emerge from the heavier-tailed filtered trace distribution, where magnitude differences between tasks are larger and the gradient toward long-competitor focus is stronger.

Finding: The W8c agent learned to attend to the longest competitor confidently (93.7%, H = 0.32–0.50) rather than collapsing to noise. This is a qualitatively different failure from W8a/b's entropy collapse: the attention mechanism is functioning correctly as an attention mechanism — it is attending to a coherent signal — but that signal is anti-SRPT under the area-integral reward on real-trace distributions. Closing this gap requires either reward shaping that directly incentivizes shortest-job-first selection at each step, a supervised warm-start on SRPT demonstrations before RL fine-tuning, or a learnable attention temperature that allows the agent to modulate its focus between shortest and longest competitors as a learned policy rather than a gradient-forced default.

5.17 F16 — Resource-Class Features Enable Round Robin Parity and Correct SRPT Selection

Adding plan_cpu and plan_mem as normalized resource-class features — expanding the per-process encoding from 3 to 5 dimensions and increasing d_attn from 8 to 16 (3,873 parameters total) — produced the first agent in this project to beat Round Robin on real Alibaba trace data.

Results on 500 filtered test episodes (master seed 42):

Agent	Mean MCT	Std	vs Round Robin	vs SRPT
SRPT oracle	15.20s	8.94s	—	0.0s
W9 AttentionDQN (5-feat)	19.22s	11.58s	**−2.09s**	+4.03s
W8c AttentionDQN (3-feat)	23.30s	10.85s	+1.53s	+7.58s
Round Robin (tier1 = 1.0s)	21.31s	14.03s	baseline	—

Four findings characterise the W9 result.

Finding 1 — First real-trace Round Robin win. W9 achieves mean MCT 19.22s, beating Round Robin by 2.09s. This is the first agent in this project to beat Round Robin on real trace data. W8c and W8d (3-feature agents, trained for 10,000 and 20,000 episodes respectively) both sat above Round Robin at 23.30s and 24.66s. The improvement came from richer state, not additional training: W8d's extended 20,000-episode run degraded SRPT agreement (39.2% vs 57.9% at 10,000 episodes for W8c), confirming the 3-feature state was at its ceiling.

Finding 2 — Identity bias corrected. The identity bias probe (candidate burst_norm = 0.20, competitor burst_norm = 0.60, identical cpu_norm = 0.125 and mem_norm = 0.339) yields Q(candidate) = −2.114, Q(competitor) = −2.664, |Δ| = 0.549, candidate preferred. This is the first correct identity bias direction on real-trace data. W8c produced |Δ| = 0.156 in the wrong direction. The resource-class features provided the signal that allowed the MLP to rank the shorter-burst candidate higher despite identical CPU and memory requests.

Finding 3 — SRPT agreement improved at every contention level. Overall agreement rose from 57.9% (W8c) to 70.5% (W9). The most dramatic improvement is at n_active = 4: 21.3% → 52.8%. Agreement at n_active = 5 remains the open problem (32.5%), consistent with the high-contention failure observed on synthetic data.

n_active	W8c agree%	W9 agree%	Δ
1	100.0%	100.0%	0.0
2	38.1%	57.5%	+19.4
3	47.1%	54.7%	+7.6
4	21.3%	52.8%	+31.5
5	—	32.5%	—

Finding 4 — Attention pattern unchanged; MLP learned to use it correctly. W9 attention still attends to the longest competitor in 84.7% of decisions at n_active ≥ 3, essentially the same as W8c (93.7%). The attention mechanism did not change its structural behaviour. What changed is that the MLP — now conditioned on cpu_norm and mem_norm in both the context and the a6 vector — learned to use the longest-competitor context as a reference for contrast rather than as a selection target. Resource-class features broke the reward degeneracy that trapped the 3-feature agent without requiring a change to the attention mechanism or the reward signal.

Honest limitations. plan_cpu and plan_mem are requested resources, not measured runtime utilisation. They encode the job's resource class at submission time and are identical for all tasks within the same job. The signal is weaker than the historical CPU usage or I/O wait ratio identified by RLScheduler and Double DQN as load-bearing. The n_active = 5 agreement ceiling at 32.5% and the 4.03s SRPT gap remain open. The 84.7% longest-competitor attention confirms that the reward signal degeneracy is not fully resolved — only compensated for at the MLP level by the additional features.

5.18 F17 — Potential-Based Reward Shaping Degrades Performance at Both Tested Scales

Potential-based reward shaping with $\Phi(s) = -\alpha \times \Sigma_i$ burst_norm_i was tested at two scales: W10A ($\alpha = 0.20$) and W10A-v2 ($\alpha = 0.05$). The shaped reward at each step is:

```
R_shaped = R_original + γ × Φ(s') − Φ(s)
```

where $\gamma = 0.99$ and R_original is the unmodified area-integral reward. Φ uses the same log-normalised burst values already in the state vector — no new observations are introduced, and the tabula rasa constraint is preserved. The shaping bonus correctly favours shortest-job completion: completing a process with burst_norm = 0.20 ($\alpha = 0.20$) yields +0.043 vs +0.004 for running the longest process one step.

Results against the W9 baseline (n = 500 filtered test episodes, greedy policy):

Agent	Mean MCT	Overall agree%	n2 agree%	tier0% at n2
W9 (no PBRS)	19.22s	70.5%	57.5%	94.5%
W10A $\alpha = 0.20$	22.15s	52.2%	46.9%	97.0%
W10A-v2 $\alpha = 0.05$	19.23s	65.7%	53.3%	95.6%

$\alpha = 0.20$ caused near-total tier0 collapse (97.0% of decisions used the smallest quantum at n_active = 2) and an 18 percentage-point SRPT agreement regression. Reducing α to 0.05 recovered mean MCT to 19.23s — statistically identical to W9 — but SRPT agreement remained 5 percentage points below W9 at every contention level. No tested α improved on W9.

Diagnosis. The shaping bonus accumulates once per decision step, regardless of which process is selected. In a discrete quantum-based scheduler, tier0 (smallest quantum = 0.25s) maximises the number of decision steps per episode and therefore maximises total accumulated shaping bonus, independent of process-selection quality. The agent at $\alpha = 0.20$ learned this: it uses tier0 in 97% of decisions at n_active = 2, a near-uniform collapse. The process-selection accuracy simultaneously collapsed because the shaping signal was dominated by quantum-choice incentives. At $\alpha = 0.05$ the incentive is weaker and MCT recovers, but the residual step-count bias still depresses agreement by 5 percentage points. PBRS in its current form is incompatible with discrete quantum scheduling without a per-quantum-size adjustment to the shaping magnitude — an additional design choice that removes the simplicity motivation for PBRS.

5.19 F18 — Learnable Attention Temperature Converges to Softer Attention, Not Sharper

A learnable inverse-temperature parameter was introduced: scores_j = $\beta \times (q \cdot K_j)$, where $\beta = \exp(\log\_\beta)$ is trained alongside the attention weights. log_β is initialised to $\log(1/\sqrt{16}) \approx -1.386$, giving $\beta_0 = 0.25$ — identical to the fixed scaling in W9. Log- parameterisation keeps β strictly positive. This adds one parameter (3,874 total vs W9's 3,873) and is otherwise architecturally identical.

Training trajectory of β:

Episode	β
0 (init)	0.250
500	0.167
1000	0.118
2000	0.076
5000	0.047
10000	0.081

β decreased monotonically from 0.250 to a minimum near 0.047 at episode 5000 before rising slightly to 0.081 at convergence — three times softer than the W9 fixed value throughout training.

Results against the W9 baseline:

Agent	Mean MCT	Overall agree%	n5 agree%	β
W9 fixed β = 0.25	19.22s	70.5%	32.5%	0.250
W10B learned β	20.63s	54.8%	10.5%	0.081

Performance regressed by 1.41s mean MCT. n_active = 5 agreement collapsed from 32.5% to 10.5%.

Diagnosis. The gradient correctly found that sharpening attention is harmful given the current 5-dim feature set. The W9 attention already concentrates 84.7% of weight on the longest competitor — a structural consequence of the area-integral reward (Section 5.16). With only burst_norm, arrived_flag, wait_norm, cpu_norm, and mem_norm in the competitor encoding, the features are insufficiently discriminative to support sharper attention at high contention: the longest-competitor signal dominates all dot products regardless of β, and higher β would simply make this concentration more extreme rather than more informative. The network's gradient identified this and drove β down to softer values where near-uniform competitor averaging is preferable to confident misattention. The n_active = 5 failure is therefore a feature- discriminability problem rather than an attention-capacity problem — the right fix is a richer competitor encoding, not a different temperature schedule.

5.20 F19 — Two-Head Attention Enables Head Specialization and Closes SRPT Gap to 3.01s

W10C replaced single-head attention (W9, d_attn = 16) with two-head attention (d_head = 8 per head, output projection W_O: 16 → 16, 4,145 parameters vs W9's 3,873). No PBRS; no learnable temperature. One architectural change only.

Results on 500 filtered test episodes (master seed 42):

Agent	Mean MCT	Std	vs Round Robin	vs SRPT
SRPT oracle	15.20s	8.94s	—	0.0s
W10C 2-head attention	18.21s	12.15s	**−3.10s**	+3.01s
W9 1-head fixed β	19.22s	11.58s	−2.09s	+4.02s
Round Robin (tier1 = 1.0s)	21.31s	14.03s	baseline	—

W10C reduces mean MCT by 1.01s and the SRPT gap from 4.02s to 3.01s — a 25% reduction.

Finding 1 — Spontaneous head specialization without supervision. The two heads diverged into qualitatively distinct roles measured across 1,064 decisions at n_active ≥ 3:

Head	Highest-attn = shortest	Highest-attn = longest	Mean w (shortest)	Mean w (longest)
Head 1 (context)	45.6%	40.9%	0.4343	0.4210
Head 2 (monitor)	14.2%	79.1%	0.3309	0.5323

Head 1 exhibits near-uniform attention across competitors — surveying the full competitor field and aggregating load context. Head 2 concentrates strongly on the longest competitor (79.1%), tracking the most threatening process as a reference signal. Neither role was imposed by supervision or architectural constraint; both emerged from the area-integral reward signal applied to the two-head architecture. This matches the hypothesis implicit in

W10B's diagnosis: single-head attention was solving two problems simultaneously — load context aggregation and competitor monitoring — with a compromise allocation. Two heads divided the labor cleanly.

Combined, either head attends to the longest competitor in 98.3% of decisions; the combined shortest-attention rate (either head) is 59.2%. The two heads together provide richer input diversity to the MLP than any single head could supply at the same total dimensionality.

Finding 2 — Largest single-step SRPT agreement improvement at n_active = 2. Agreement at n_active = 2 increased from 57.5% (W9) to 69.9% — a 12.4 percentage point gain, the largest improvement at any single contention level across all experiments in this project.

n_active	W9 agree%	W10C agree%	Δ
1	100.0%	100.0%	0.0
2	57.5%	69.9%	+12.4
3	54.7%	54.6%	−0.1
4	52.8%	38.8%	−14.0
5	32.5%	30.3%	−2.2

Finding 3 — High-contention ceiling persists. Tier0 collapse at n_active ≥ 4 (96–98%) and the n_active = 5 agreement floor (30.3%) remain. Two-head attention improved moderate-contention performance but did not address the feature-discriminability ceiling at high contention. The W10B diagnosis stands: five planned resource features are insufficient to support informative attention at n_active = 5, regardless of how many heads process them.

Finding 4 — Permutation invariance preserved exactly. Unit test: $|Q\_A - Q\_B| = 2.78 \times 10^{-16}$ (machine epsilon). The output projection W_O operates on the concatenated head outputs, which are each independently permutation-invariant over the competitor set. The concatenation and linear projection preserve this property.

Interpretation. Single-head attention at d_attn = 16 was attempting to perform both competitor monitoring and load context aggregation with a single set of query-key-value weights. The W10B experiment confirmed that the gradient identified this compromise: β converged to 0.081 (softer than the W9 fixed value) because the single head could not sharpen without making its longest-competitor focus more extreme and losing load context. Two heads removed the trade-off by assigning each role a dedicated weight subspace. The MLP receives a 16-dim context vector that simultaneously encodes both the load landscape (Head 1) and the maximum-threat reference (Head 2), enabling correct SRPT-aligned selection at moderate contention.

Extended training to 20,000 episodes (W10C-ext) triggered the stop gate at episode 15,000 — MCT rose from 19.32s to 19.60s. Final evaluation produced 20.32s mean MCT, a 2.11s regression from W10C at 10,000 episodes. Head specialization partially collapsed: Head 1 drifted from near-uniform context attention (45.6% shortest / 40.9% longest) toward longest-monitoring (22.1% / 76.3%), mirroring the W8c/W8d degradation pattern. The 10,000-episode ceiling is consistent across both the 1-head (W9/W8c) and 2-head (W10C) architectures, confirming it is a dataset and feature-set constraint rather than an architecture-specific artifact.

5.21 F20 — Value-Curve Reward Without Observable Curve Parameters Produces No Learning

W10C-VC1 replaced the area-integral reward with a value-curve objective. At each step the reward is `sum(-value_delta(p) for p in runnable) / 20.0`, where `value_delta(tau, floor, base, wait, q) = base × max(floor, exp(-(wait+q)/tau)) - base × max(floor, exp(-wait/tau))`. Processes are divided into steep curves (tau ∈ [3,10]s, floor=0.2, sampled with probability 0.5) and smooth curves (tau ∈ [40,100]s, floor=0.0). Tau and floor are sampled per episode but were not added to the state vector; the per-process encoding remained the 5-dim W10C representation (burst_norm, arrived_flag, wait_norm, cpu_norm, mem_norm).

The stop gate fired at episode 2000: mean MCT across three consecutive checkpoints was 22.34s → 22.74s → 23.29s, all above the 18.21s W10C baseline. Final evaluation of the episode 2000 checkpoint: mean MCT = 26.56s, versus Round Robin (21.31s, regression +5.25s) and W10C (18.21s, regression +8.35s). Steep SRPT agreement: 30.5%; smooth SRPT agreement: 31.0%; gap = −0.58pp — near random and in the wrong direction. Head specialization analysis across 1,064 decisions at n_active ≥ 3 showed no curve-type dependence: Head 1 attended to the shortest competitor in 52.8% / 42.8% of steep / smooth decisions; Head 2 to the longest in 59.2% / 63.8%. The two-head specialization pattern from W10C reproduced unchanged regardless of whether the SRPT process carried a steep or smooth curve.

The identity bias probe, run across all four curve-type combinations (steep candidate / steep competitor, steep / smooth, smooth / steep, smooth / smooth), returned Δ = +0.035765 in every case — identical to six significant figures. The Q-value delta the network assigns to the candidate over the competitor is independent of either process's curve type. This is the diagnostic signature of non-stationary Q-values: identical observations produced different rewards depending on the sampled tau and floor values, making the Bellman target a moving average of outcomes from qualitatively different reward regimes rather than a convergent signal. No process selection aligned with urgency was learnable under these conditions.

Finding: When the reward depends on unobserved process properties, the identity bias probe returns identical Δ across all curve-type combinations. This is a cleaner diagnostic of reward non-stationarity than the loss curve or MCT trend alone: it confirms that the network produces no urgency-differentiated output regardless of training duration or architecture quality. The fix is observability, not architecture.

5.22 F21 — Value-Curve Oracle Cannot Beat Round Robin on MCT: Structural Divergence

Before training with extended state, a greedy value-curve oracle was diagnosed on 50 episodes (seed=42). The oracle scores each (process, quantum) pair and selects the maximum. The initial scoring formula was `score = -value_delta(tau, floor, base, wait, q) / q` — a marginal value rate. This formula analytically always maximises at the smallest quantum: dividing by q makes the score a per-second rate, which a 0.5s quantum can always exceed because it is cheaper per unit time regardless of value profile. The 50-episode diagnostic confirmed: 100% tier-0 (0.5s) selection across all decisions, steep and smooth alike. Oracle MCT = 23.27s versus Round Robin = 21.61s (+1.66s worse). A scoring formula that selects the minimum quantum at every step is not a meaningful oracle.

The formula was corrected to `score = −value_delta(tau, floor, base, wait, q)` — total value saved by the action, with no per-second normalisation. Larger quanta can now win if they save more total value. Re-running the 50-episode diagnostic with the corrected formula: tier-0 collapse resolved. Steep processes (tau < 15s): 79.6% short quantum (0.5s), 2.4% medium (2.0s), 18.0% long (8.0s). Smooth processes (tau ≥ 15s): 0% short, 0% medium, 100% long. Steep SRPT agreement: 81.9%; smooth SRPT agreement: 57.9%; gap: +24.0 percentage points. The oracle correctly identifies that steep processes — near their decay cliff — benefit from short, frequent quanta, while smooth processes — decaying negligibly over any feasible quantum — receive the maximum quantum to accumulate the largest total value saved per step. This is the expected behaviour of a value-aware greedy policy. Oracle MCT at 50 episodes: 22.58s versus Round Robin 21.31s (+0.98s). At 500 episodes (master seed 42): oracle MCT = 21.86s, std = 13.00s — still above Round Robin.

The +0.98s gap at the oracle level is a structural result, not sampling noise. Smooth processes under a value-curve objective receive 100% long quanta. Under SRPT, those same processes would be preempted immediately in favour of any shorter remaining-burst competitor. The value-curve oracle trades SRPT-aligned preemptions for long smooth-process quanta that maximise value preservation. MCT increases as a consequence: every long smooth-process quantum holds the CPU away from shorter competitors for 8 seconds at a time. No amount of state extension or learning can close this gap — the oracle itself cannot close it.

Finding: The value-curve oracle cannot beat Round Robin on MCT even with perfect greedy scoring. The divergence is structural: smooth processes receive 100% long quanta to maximise value preservation, violating the SRPT preemption that minimises MCT. This tension is confirmed before any learning occurs and bounds the MCT performance achievable under value-curve reward regardless of agent architecture.

5.23 Quantum Selection Impossibility: Ablation Results

The Lemma in Section 3.1.3 predicts that any per-task quantum scoring function must degenerate to a boundary selection (always q_min or always q_max). Three ablations test this prediction on the Alibaba-derived trace with calibrated value curve parameters (τ_steep = 900s, τ_smooth = 800s).

quantum_only — Round Robin process selection, quantum selected by f_rate(q) = ΔV/q:

Policy	q_min % (steep)	q_min % (smooth)	q_med %	q_max %	total_value
quantum_only	100.0%	100.0%	0.0%	0.0%	250.55
quantum_only_fixed	0.0%	0.0%	0.0%	100.0%	159.21
ordering_only	—	—	100.0%	—	15.37
value_aware (full)	mixed	mixed	mixed	mixed	250.55

quantum_only selects q_min at 100% frequency for both steep and smooth task types, for both default (τ = 25/100) and calibrated (τ = 900/800) initializations. The result is parameter-independent: the monotonicity of f_rate holds for all τ > 0, so no calibration can break the degeneracy.

quantum_only_fixed uses f_total(q) = ΔV (no per-unit-time normalisation). It selects q_max at 100% frequency — the opposite degeneracy predicted by the Lemma. Total value = 159.21, below quantum_only (250.55), because large quanta hold the CPU away from other waiting tasks for extended periods, accumulating delay penalties that outweigh the per-task value preservation gain.

ordering_only uses gradient ordering (argmax_i |∇Φ_i|) with a fixed q_med quantum. Total value = 15.37 — a collapse to near-zero. The cause is smooth task starvation: steep tasks (smaller τ, larger |∇Φ|) are selected preferentially, while smooth tasks accumulate delay until their gradients surpass the steep tasks at plateau. By that point, smooth task delays exceed the epoch, and they complete with near-zero value remaining. The ordering is locally correct but globally catastrophic without quantum selection to compensate.

Finding: All three per-task variants degenerate as predicted. The full value_aware policy — which uses queue-global state to inform both ordering and quantum selection — achieves total_value = 250.55, equal to quantum_only (which achieves high total_value through a different mechanism: frequent preemption via small quanta). The distinction is fairness: quantum_only achieves equal total_value through RR ordering that does not discriminate by urgency; value_aware achieves it through urgency-sensitive ordering confirmed by the Corollary's queue-global requirement.

5.24 F22 — Observable Curve Parameters Produce Urgency-Differentiated Behavior at MCT Cost

W10C-VC2 extended the per-process feature vector from 5 to 7 dimensions, inserting `tau_norm = p.tau / 100.0` at position [3] and `floor = p.floor` at position [4] and shifting cpu_norm and mem_norm to positions [5] and [6]. State dimensionality: 5 × 7 = 35. Network parameter count: 4,369 (W_Q/K/V: 7×8 per head instead of 5×8; MLP layer 0: 24×64). The oracle scoring formula uses `−value_delta` with no `/q` denominator. He initialisation; W10C weights not loaded. Gradient check: max relative error 2.53×10^{-7} at h0_W_Q[29], passed (threshold 1×10^{-4}). Full 10,000-episode run; stop gate (total_value trending down for 3 consecutive checkpoints) did not fire.

Primary metrics on 500 filtered test episodes:

Checkpoint	total_value	value_rate (s^{-1})	steep_vp%	smooth_vp%	MCT	vs RR
ep 2000	−0.1426 ± 0.0605	−0.003408	62.60%	84.52%	22.90s	+1.59s
ep 5000	−0.1404 ± 0.0565	−0.003369	62.60%	84.73%	21.46s	+0.15s
ep 10000	−0.1554 ± 0.0698	−0.003666	53.76%	83.42%	23.76s	+2.45s

Best checkpoint: episode 5000 on both total_value and MCT. Smooth value preserved (84.52–84.73%) consistently exceeds steep value preserved (53.76–62.60%), consistent with the oracle's differential quantum treatment: long quanta preserve smooth-curve value; short quanta are needed for steep curves near their decay cliff. The SRPT agreement gap between steep and smooth processes widened monotonically: 5.83pp at episode 2000, 9.7pp at episode 5000, 10.13pp at episode 10000. Direction is correct — the agent shows stronger SRPT agreement for steep (urgent) processes throughout training. The gap widened despite absolute agreement declining at episode 10000, confirming the directional divergence is stable even as overall policy quality degrades late in training.

Finding 1 — Urgency differentiation confirmed. The identity bias probe returned qualitatively distinct results across the four curve-type combinations: steep/steep $\Delta = +0.035639$ (candidate preferred, SRPT-correct); steep/smooth $\Delta = -0.095839$ (competitor preferred, SRPT-incorrect); smooth/steep $\Delta = -0.036831$ (competitor preferred, SRPT-incorrect); smooth/smooth $\Delta = -0.003082$ (competitor preferred, SRPT-incorrect). Unlike VC1, where all four combinations returned $\Delta = +0.035765$, the VC2 network produces curve-type-dependent Q-values. Tau_norm and floor are read and used. The network assigns higher Q-value to smooth competitors than to steep candidates in three of four combinations — an inverted preference. The cause is the value-curve reward itself: a smooth process has more value remaining at any wait time (slow decay, high floor) and therefore generates a larger `-value_delta` magnitude per step. The agent learned that smooth processes offer larger reward and preferentially schedules them, the opposite of urgency-first behaviour. The correct feature dependency (tau and floor matter) is learned. The correct preference ordering (steep first, to preserve rapidly decaying value) is not.

Finding 2 — MCT cost is structural, not a training failure. VC2's best MCT (21.46s at episode 5000) is 3.25s worse than W10C (18.21s) and 0.15s worse than Round Robin. This gap is not addressable by longer training or architecture modification. The oracle diagnostic (F21) established that even a perfectly greedy value-curve policy cannot beat Round Robin on MCT, because smooth processes receive long quanta that block shorter competitors. The VC2 agent's 0.15s margin over Round Robin at episode 5000 represents the tightest MCT feasible under the value-curve objective without a reward term that explicitly penalises MCT delay.

5.25 F23 — Reward Signal Design Determines Learnable Objectives

The W10C-VC1 and W10C-VC2 experiments jointly establish a finding that performance tables cannot convey: reward signal design determines which structural behaviours an agent is capable of producing, independent of architecture quality. W10C (area-integral reward) beats Round Robin by 3.10s on MCT and achieves 25% reduction in the SRPT gap over W9. It cannot, by construction, differentiate between steep and smooth processes — both contribute identically to the n_active × q_actual penalty. The identity bias probe across the four curve-type combinations would return identical Δ for any MCT-trained agent, because the reward provides no gradient signal that depends on tau or floor. W10C-VC2 (value-curve reward, tau_norm and floor observable) produces curve-type-dependent Q-values, a widening steep/smooth SRPT agreement gap, and differential value preservation (84.73% smooth vs 62.60% steep at the best checkpoint) — none of which a W10C-class agent can produce regardless of training budget or head count.

The cost is structural and was confirmed before training. The greedy value-curve oracle — which makes optimal per-step decisions given the value objective — achieves MCT = 21.86s at 500 episodes, +0.55s above Round

Robin. The oracle assigns 100% long quanta to smooth processes to maximise value preservation; SRPT would preempt those same processes in favour of shorter competitors. These objectives are not two approximations of the same target at different performance levels. They are different targets. An agent trained on value-curve reward and evaluated on MCT is being scored on a metric its reward signal was not designed to optimise. The VC2 best checkpoint at 21.46s — within 0.15s of Round Robin — sits near the MCT boundary accessible to any agent constrained by the value-curve objective.

The widening steep/smooth SRPT gap (5.83pp → 10.13pp across 10,000 episodes) confirms that the urgency signal in tau_norm and floor is being learned progressively. The inverted preference in the identity bias probe (smooth competitor preferred over steep candidate in three of four combinations) identifies what is not yet learned: that preserving steep-curve value requires scheduling steep processes *first*, not last. Closing this gap would require either a reward term that explicitly penalises late scheduling of steep processes relative to their tau (a deadline-aware shaping bonus, analogous to the PBRS analysis in F17 but curve-type-conditioned) or a secondary supervision signal that labels steep-first orderings as preferred. Neither is achievable under the tabula rasa constraint as defined.

Conclusion: Reward signal design determines which objective an RL scheduler optimises. Value-curve reward produces urgency-differentiated behaviour (steep/smooth Q-value divergence, widening SRPT gap) that uniform reward cannot produce. This comes at measurable MCT cost because the two objectives structurally diverge for slow-decaying processes — a tension confirmed at the oracle level before any learning occurs.

5.26 Gradient Alignment and Tau Calibration

The gradient ordering described in Section 3.1.1 requires that co-queued tasks have meaningfully different |∇Φ_i|. Whether this condition holds depends critically on the relationship between τ and the empirical delay distribution on the target trace.

On the Alibaba 2018 batch_task trace (filtered, p75 cutoff), task delays follow a heavy-tailed distribution. Measured median delays are: steep tasks, 5,279s; smooth tasks, 38,665s. The default τ values inherited from the synthetic toy environment are τ_steep = 25s and τ_smooth = 100s — three to four orders of magnitude below the actual delay scale. At these parameters, exp(−5,279/25) ≈ 10^{−92}: the gradient |∇Φ_i| = (base/τ) · exp(−d/τ) is numerically zero for every task in the queue. The ordering is undefined — all tasks appear equally urgent under the wrong τ scale.

Tau calibration. A principled initialization sets τ such that V(median_delay) = V(0)/2 — i.e., the half-life of the value curve matches the empirical median delay. This gives τ_calibrated = median_delay / ln(2). Applied to the measured medians: τ_steep = 5,279/0.693 ≈ 7,617s. However, using the p25 delay for steep tasks (which cluster at lower delays) gives a more conservative τ_steep = 637/0.693 ≈ 900s. For smooth tasks, the median gives τ_smooth = 38,665/0.693 / 48 ≈ 800s (where the 48× factor accounts for the smooth-task epoch scale). The approved calibrated values are τ_steep = 900s, τ_smooth = 800s.

Near-tie result. Even with calibrated τ, 93–100% of ordering decisions are near-ties (top-2 gradients within 10%) on the batch-dominated trace. The cause is structural: 92% of tasks are smooth (τ_smooth ≈ 800s), and smooth tasks co-queued with similar delays have gradients differing by exp(−ε/τ_smooth) → 1 as τ_smooth grows relative to inter-task delay differences. Tau calibration changes the numerical scale but not the near-tie fraction, because it raises τ for all tasks simultaneously, preserving the ratio. The near-tie problem is trace-composition-determined — it is a consequence of the 92%/8% smooth/steep split, not a calibration failure.

Finding: Default τ values produce zero gradients on real trace delays; calibrated τ values produce nonzero but near-identical gradients on batch-dominated traces. Gradient-based ordering is informationally void on this workload composition regardless of τ calibration.

5.27 Fairness Analysis

5.27.1 Metric Definitions and Limitations

Standard fairness metrics for scheduling evaluate completion time distributions. **Jain's Fairness Index** (JFI) measures completion time uniformity: JFI = $(\Sigma_i x_i)^2 / (n \cdot \Sigma_i x_i^2)$, where x_i is the completion time of task i. JFI = 1 for perfect equality; JFI → 1/n for maximum inequality. **Slowdown Variance** (SDV) measures the variance of per-task slowdown (completion time / burst length), capturing proportional unfairness.

Both metrics share a fundamental limitation: they are blind to value curve heterogeneity. A schedule that completes all steep tasks and all smooth tasks at the same absolute delay d achieves JFI = 1 and SDV = 0 — but a steep task at delay d has preserved far less value ($\exp(-d/\tau_{steep})$) than a smooth task ($\exp(-d/\tau_{smooth})$). The schedule is metrically fair but value-unfair.

5.27.2 Value-Rate Fairness Index (VRFI)

We define a new fairness metric that captures value-curve-adjusted fairness. The **Value Loss Rate** (VLR) for task i is:

```
VLR_i = (V_i(0) − V_i(delay_i)) / delay_i
```

VLR_i measures value lost per unit time during the task's wait. For non-plateau tasks, VLR_i → base_i/τ_i as delay_i → 0 (maximum loss rate). At plateau, VLR_i = (base_i − floor_i) / delay_i (decreasing with delay, reflecting sunk cost).

The **Value-Rate Fairness Index** is:

```
VRFI = 1 − std(VLR) / mean(VLR)
```

VRFI = 1 − CV(VLR), where CV is the coefficient of variation. VRFI = 1 for perfect VLR equality (all tasks lose value at the same rate); VRFI → −∞ as dispersion grows. The metric is negative for all preemptive policies on mixed steep/smooth workloads: steep and smooth tasks have structurally different base/τ ratios, so their VLR distributions do not overlap.

Boundary verification. Five synthetic examples confirm the metric's behaviour: (A) identical tasks, same delay: VRFI = 1.0. (B) identical tasks, varying delay: VRFI < 1, decreasing as delay spread increases. (C) extreme inequality (one task completes instantly): VRFI → −∞. (D, key falsification) same delay, mixed profiles: Jain's = 1.0, SDV = 0, VRFI = 0.641 — JFI and SDV cannot distinguish value-fair from value-unfair schedules. (E) negative-mean VLR (impossible under standard value curves): undefined by convention.

5.27.3 Five-Policy Fairness Results

Policy	JFI global	SDV global	Starvation (%)	VRFI global	VRFI smooth	VRFI steep	Rank sum
value_aware	0.961	566	0.0%	−8.063	0.749	0.612	17
mlfq	0.943	1,204	0.0%	−9.241	0.783	0.541	19
rr	0.978	15,819	0.0%	−7.914	0.926	0.488	24
cfs	0.931	8,732	2.1%	−10.033	0.801	0.499	28
fcfs	0.874	22,441	9.0%	−11.820	0.712	0.381	32

Starvation threshold: slowdown > 3× group-median slowdown. Rank sum computed over all 8 metric-by-group cells; lower is better.

value_aware achieves the lowest rank sum (17 of 40 possible) and zero starvation. Its SDV (566) is 28× lower than Round Robin (15,819), indicating substantially lower slowdown variance despite having a worse global

VRFI than RR. The within-group VRFI reveals why: smooth VRFI (0.749) is competitive but below RR (0.926), while steep VRFI (0.612) is the second-highest. The global VRFI is pulled negative by the structural base/τ divergence between steep and smooth populations — a trace property, not a policy failure.

fcfs produces 9% starvation and the worst scores across all metrics. **cfs** produces 2.1% starvation despite its proportional-fairness design, because the Alibaba trace's heavy-tailed duration distribution overwhelms the virtual runtime balancing on short scheduling horizons.

Finding: JFI and SDV are insufficient for value-curve workloads — Example D demonstrates a schedule where both metrics report perfect fairness while VRFI correctly identifies value-rate disparity. The VRFI rank ordering (value_aware best, fcfs worst) is consistent with the SDV ordering for 4 of 5 policies but inverts for rr (SDV worst among surviving policies; VRFI third-best globally). The inversion reflects RR's high-SDV, low-VLR-disparity profile: RR distributes CPU time uniformly, producing similar VLR across task types at the cost of high absolute slowdown variance.

5.28 Value Curve Parameter Optimization

Given the tau calibration finding (Section 5.26) and the degeneracy results (Section 5.23), a natural question is whether gradient descent on the value curve parameters (τ_steep, floor_steep, base_steep, τ_smooth, base_smooth) can recover a parameter configuration that avoids both the near-tie problem and the quantum degeneracy.

Two loss functions were evaluated: **Loss A** (efficiency): Σ_i (V_i(0) − V_i(delay_i)) — total value lost across the trace. **Loss B** (fairness): Var(VLR) — variance of value loss rates, targeting VRFI maximisation.

5.28.1 Degeneracy Hierarchy Under Loss A

Loss A admits three exploits in decreasing order of gradient magnitude:

1. **Floor collapse**: raising floor_steep → V_i(0) ≈ V_i(delay_i) for all i → total value lost → 0. At both default (τ = 25/100) and calibrated (τ = 900/800) initializations, gradient descent converges to floor_steep = 0.94 regardless of initialization, closing the value gap by making the curve nearly flat.

2. **Base shrink**: after floor is anchored via regularization ($\lambda \cdot (\text{floor} - 0.2)^2$), base_steep and base_smooth shrink toward 0 → V_i(0) → 0 → loss → 0. At λ = 1: floor_steep = 0.895 (exploit persists). At λ = 10: floor_steep = 0.234 (anchored), base_steep = 0.03, base_smooth = 0.03 (new exploit active). At λ = 100: floor_steep = 0.203, base values < 0.05.

3. **Tau drift**: with base and floor fixed at their true values (base = 1.0, floor = 0.2), only τ parameters are free. 72% of steep tasks are at plateau (gradient = 0); tau_smooth receives zero gradient because smooth task delays (38,665s median) >> τ_smooth (800s) → gradient term exp(−delay/τ) ≈ 0. Over 1,000 gradient steps: τ_steep moves 900 → 910s (Δ = +10s), loss reduction = 0.013 (0.004% of total loss). Tau optimization is legitimate but informationally starved.

5.28.2 Loss B: Fairness as Exploit-Resistant Objective

Loss B (Var(VLR)) resists all three Loss A exploits: floor collapse increases VLR disparity (steep plateau VLR = (base−floor)/delay diverges from smooth VLR); base shrink reduces mean VLR but increases CV; tau drift at the calibrated scale produces a genuine gradient signal. With calibrated initialization (τ_steep = 900, τ_smooth = 800), Loss B achieves VRFI = +0.289 after 1,000 gradient steps — the first positive VRFI on this trace across any policy or parameter configuration. This is the only optimization objective that improves rather than degrades the fairness metric.

Finding: Efficiency-based parameter optimization (Loss A) degenerates regardless of regularization, exposing successive exploits in the value curve parameterization. VRFI-based optimization (Loss B) is exploit-resistant and achieves positive within-group VRFI, at the cost of not directly optimizing task throughput. Future work should investigate joint optimization of Loss A + Loss B under fixed-base, fixed-floor constraints, where the only free parameters are the τ values — the single axis where a legitimate gradient signal survives.

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6. Discussion

6.1 What the Agent Discovered

The Week 5 agent learned, from reward alone and without supervision:

- **Perfect masking compliance.** Never selects a completed or unarrived process. This was not imposed by reward shaping — invalid actions are simply excluded from argmax. The agent learns to route its probability mass away from them during training.

- **Quantum strategy matching SRPT.** At n_active ≥ 2, the agent uses 1ms quanta in >99% of decisions. SRPT uses 1ms exclusively. The convergence is not coincidental: 1ms quanta allow the most frequent preemption decisions, giving the agent the highest degree of control over process ordering. The reward signal — which penalises every millisecond a process remains active — implicitly favours fine-grained control.

- **Partial SRPT process selection.** At n_active = 2, the agent agrees with SRPT 74.3% of the time. For the two-process case, the action-conditioned invariance guarantee is maximally effective: one candidate, one competitor, identical PID-masked context. The agent has effectively learned to prefer shorter remaining bursts in the binary case.

- **Near-optimal OOD performance.** On the fixed OOD set (bursts 4/25/2/50/8ms, three simultaneous arrivals at t=0), the agent achieves 29.60ms against SRPT's 28.4ms — a gap of 1.20ms, the closest any agent in this project has come to the optimum on that configuration.

What it did not discover: a correct process selection rule for three or more simultaneous candidates (65–68% accuracy, plateaued), and full permutation invariance over competitor features (residual $|\Delta| = 0.051$ in the bias probe).

6.2 The Permutation Invariance Ceiling

The process-selection ceiling at 65–68% for n_active ≥ 3 was not a training-time phenomenon in Week 5. The loss converged to 0.0001. The ceiling was architectural.

The action-conditioned design (W5) shares weights across candidate PIDs — a query for P0 uses the same network as a query for P4. But the competitor context (s_masked) still presents the four non-candidate processes at their PID-indexed positions. A state where P1, P2, P3, P4 are runnable competitors and P0 is the candidate produces a different s_masked than a state where P0, P2, P3, P4 are competitors and P1 is the candidate, even if the competitor burst magnitudes are identical — because the burst values appear at different positions in the 15-dim vector.

Week 7 addressed this directly. AttentionDQN replaces the flat competitor concatenation with dot-product attention over competitor key-value pairs. The softmax operation produces identical context vectors for any permutation of the competitor inputs, reducing the effective number of distinct input configurations from 120 to 1 for a fixed set of burst magnitudes. Permutation invariance unit test: $|\Delta| = 0.000$ (W7) vs $|\Delta| = 0.0335$ (W5). The architectural ceiling from W5 was closed.

However, full invariance did not close the performance ceiling at high n_active. The W7 agreement at n_active = 3 improved substantially (90.9% vs W5's 65.4%), but collapsed at n_active = 5 (40.7% vs W5's 71.9%). The explanation is the residual entropy ceiling discussed in Section 5.8: with $H \approx 0.62$ at termination, the attention over 4 competitors is near-uniform, and the context vector degenerates toward mean-pooling. A fixed-temperature softmax cannot sharpen beyond the entropy floor imposed by the annealing schedule.

The remaining open problem is not architectural but parametric: a learnable inverse-temperature parameter β (scaling dot-product scores as $\beta/\sqrt{d}$ rather than $1/\sqrt{d}$, with β trained alongside W_Q, W_K, W_V) would allow the attention to sharpen beyond the annealing floor. This is a one-parameter change that retains the full permutation invariance guarantee while removing the fixed-temperature constraint that causes n_active = 5 failure.

A second open problem emerged from the real-trace experiments: **reward signal ambiguity on heavy-tailed distributions**. The area-integral reward (`−n_active × q_actual`) is theoretically equivalent to minimising mean turnaround time by Little's Law, and it successfully drives SRPT approximation on the synthetic uniform distribution. On the filtered Alibaba trace, however, the same reward creates gradient pressure toward attending to the longest competitor — a locally coherent strategy that is globally anti-SRPT. The agent learns that the longest task is the most informative context signal and focuses attention on it accordingly, but does not learn to *schedule* the shortest task first as a consequence. On synthetic data, this failure did not emerge because the uniform distribution's shallower magnitude spread produced a different gradient landscape. On real data with p95/p50 = 7.2× (filtered) or 39.7× (full trace), the longest task exerts disproportionate gradient influence and the attention converges to a "monitor the biggest threat" strategy rather than a "identify the shortest job" strategy. Dense reward shaping that provides a per-step signal for SRPT-aligned selections — or SRPT-supervised pre-training as a warm-start before RL fine-tuning — may be required to break this degeneracy on real workloads.

6.3 The Tabula Rasa Constraint and Its Cost

The state vector used throughout this work — remaining burst, arrival flag, wait time, three features per process — was not chosen for convenience. It was chosen as a constraint. The deliberate premise of this experiment is that no domain knowledge should enter the agent's observation: no job type priors, no I/O wait history, no CPU utilisation rolling average, no priority class. If the reward signal alone is sufficient to discover scheduling policy, the state should carry only what a bare process descriptor contains.

The constraint enabled three things that would otherwise be confounded. First, it produced a clean architectural progression: each failure mode was diagnosable in isolation because the observation space held fixed. Identity bias, discretisation collapse, and attention entropy collapse were architectural failures, not feature failures — a distinction that disappears when the state vector changes between experiments. Second, the permutation invariance analysis is exact precisely because the three-feature encoding has a known, regular structure: reshaped as (N_PROCESSES, 3), it admits a formal proof that masking and attention preserve order-invariance over the candidate slot. A heterogeneous feature vector with variable-length job-type encodings would break this argument without a more complex treatment. Third, the failure modes themselves — in particular, the anti-SRPT longest-competitor attention (Section 5.15) and the reward degeneracy on heavy-tailed distributions (Section 6.2) — are precisely diagnosable because the state provides no confound. The agent failed not because it lacked I/O information but because the reward signal created gradient pressure in the wrong direction. That diagnosis would be harder to make cleanly if the state included features that partly compensate for the reward's ambiguity.

The cost is the real-trace performance ceiling. On filtered Alibaba trace data, the W8c and W8d agents (3-feature state) achieved mean MCT above Round Robin (23.30s and 24.66s versus 21.77s) — not crossing the Round Robin threshold despite one agent receiving double the training episodes. Extended training (W8d, 20,000 episodes) degraded SRPT agreement from 57.9% to 39.2%, confirming the 3-feature state was at its ceiling rather than undertrained. The W9 agent, which adds plan_cpu and plan_mem as the two proxy features available in the Alibaba trace, crosses the Round Robin threshold for the first time at 19.22s (−2.09s vs RR). The literature's evidence points directly at what is missing. Across RLScheduler, SCHED, and Double DQN, two features appear consistently as load-bearing: I/O wait ratio (the fraction of recent time a process spent blocked on

I/O, which separates CPU-bound from I/O-bound tasks) and historical CPU usage (a rolling estimate of actual CPU consumption, which provides a prior on remaining burst that does not require direct burst observability). Neither appears in the current state vector.

Adding these features is a concrete architectural change. The per-process feature vector grows from 3 dimensions (remaining burst, arrival flag, wait time) to 5 (adding I/O wait ratio and historical CPU usage estimate), expanding the state vector from 15 to 25 dimensions. The attention module's query and key projections — currently mapping 3 → 8 — would need to absorb the wider input; a doubling of d_attn from 8 to 16 is a natural choice, adding approximately 480 parameters to the 3,041-parameter current model. The normalization scheme requires separate treatment: burst is normalised via log1p against BURST_P95; I/O wait ratio is already in [0, 1] and requires no scaling; historical CPU usage carries a different magnitude and temporal structure and likely requires its own rolling-percentile normalization rather than a fixed denominator. The replay buffer schema, the state encoding function, and the environment's process tracking all require modification to supply the two new features. None of these changes are architecturally novel. Each is a standard extension with a known implementation path.

Week 9 confirmed this: adding plan_cpu and plan_mem as resource-class proxies, despite encoding requested rather than measured usage, was sufficient to beat Round Robin and correct identity bias direction on real trace data. Week 10C confirmed and extended this: two-head attention reduced mean MCT by 1.01s and SRPT agreement at n_active = 2 by 12.4 percentage points, with spontaneous head specialization into context and selection roles — but the high-contention ceiling (n_active ≥ 4) remained, confirming that planned rather than measured resource features are the binding constraint at five-process contention.

6.4 Why the Spurious Quantum Discovery Matters

The Week 3 load-adaptive quantum pattern is worth examining beyond its status as an artefact. The pattern was: - Stable and reproducible across evaluation runs. - Load-correlated in a directionally plausible way (larger quanta as more processes compete). - Coherent enough to generate a credible post-hoc explanation grounded in queuing theory.

A practitioner examining the Week 3 agent without the Week 5 ablation might report a genuine discovery: "the agent learned to amortise context-switch costs under high load, a behaviour not explicit in the reward signal." That paper would be wrong.

The source of the error is that PID-indexed action heads accumulate asymmetric gradient histories across training episodes. If high n_active situations tend to involve process set configurations where certain PIDs are active — and in this environment they do, because arrival ordering determines which PIDs are present — then the gradient pressure toward particular quanta in those PID-indexed neurons looks, from the outside, like load-sensitive behaviour.

The generalised lesson: any RL agent with position-indexed action heads can produce policies that are coherent, reproducible, and superficially interpretable while being positional artefacts. The diagnostic is simple — an architecture ablation that shares weights across positions — but requires knowing to run it. Policy analysis without an invariance ablation is incomplete.

6.5 Limitations

Three limitations of this work are worth stating explicitly, as they bound the scope of the conclusions.

N = 5 is a toy, and scaling changes the architecture non-trivially. The fixed N = 5 with a known arrival slot set was chosen for diagnostic clarity, not representativeness. Scaling to variable N requires a dynamic candidate pool, which in turn requires the attention-based competitor aggregation identified as the next architecture step. This is not a drop-in change: it modifies the input dimensionality, the batching strategy, and the replay buffer schema. The performance conclusions reported here — the 8.38ms SRPT gap, the 74.3% two-process agreement — do not extrapolate to larger N without retraining and re-evaluation.

U[1, 60]ms burst distribution is not representative of real workloads. Real CPU burst distributions are heavy-tailed and multi-modal, reflecting a mix of short interactive tasks and long batch jobs. A uniform distribution over [1, 60]ms covers neither the sub-millisecond interactive regime nor the multi-second batch regime that dominate production schedulers. An agent trained on U[1, 60]ms may learn policies that are well-calibrated for mid-range bursts but fail at the tails — and the tails are where scheduling decisions matter most.

10,000 training episodes on a fixed arrival slot structure may have overfit the arrival distribution. Arrival times are drawn from a fixed set {0, 2, 5, 8, 10}ms across all episodes. An agent trained on this distribution may implicitly learn the arrival timing rather than general scheduling principles. A holdout evaluation using arrival times sampled from a continuous distribution outside this set would test whether the state encoding (which includes wait_time/300) generalises beyond the training arrival structure.

6.6 Connection to Real Scheduler Traces

The identity bias analysis methodology — probe states that isolate a single scheduling decision, permutation invariance unit tests that verify architectural guarantees hold in practice — is a diagnostic technique applicable to any RL scheduler regardless of environment scale. It is the most directly transferable contribution of this work. Any scheduler trained with position-indexed action heads should be subjected to an invariance ablation before its policy patterns are reported as learned behaviour.

What else would likely transfer: the reward formulation (area integral of active count) maps directly to a real latency objective if process state is tracked at each scheduling decision. The action-conditioned architecture, generalised to variable N via attention-based competitor aggregation, produces a policy that is robust to process ordering by construction — a property essential in any real workload where the same task mix can arrive in any order.

What would not transfer without modification: remaining burst observability (requires estimation from hardware counters), clean episode boundaries (requires a non-episodic reward formulation for continuously-arriving workloads), and the fixed-N assumption (requires variable-length candidate pools with the attention mechanism needed for full permutation invariance).

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7. Conclusion

Over eight weeks, without expert demonstrations, handcrafted features, or a target policy, a tabula rasa RL agent progressed from a tabular scheduler with zero cross-episode variance through four increasingly capable architectures, closing the gap to the preemptive optimal from 17.34ms (Week 3) to 8.38ms (Week 5) to a different quality frontier: an AttentionDQN (Week 7) that achieves lower variance than SRPT itself (std = 21.82ms vs 23.96ms) while meeting the 80% SRPT-agreement target at $n\_active \le 3$ for the first time.

Eight failure modes were identified, diagnosed, and resolved or explicitly left open. The honest negative results — discretisation collapse (W1), identity bias worsening with the move to DQN (W3), a spurious quantum artefact indistinguishable from a learned heuristic without ablation (W3/W4), attention entropy collapse before TD signal accumulates (W7), $n\_active = 5$ failure from residual entropy (W7), linear-normalization failure on heavy-tailed real traces (W8), attention entropy collapse on the full Alibaba trace (W8), anti-SRPT longest-competitor attention on the filtered trace (W8c) — are as central to the contribution as the positive ones.

Three formal results emerged from the value-curve analysis. The Quantum Scoring Monotonicity Lemma (Section 3.1.3) proves that no per-task scoring function can produce mixed quantum selection for any $\tau > 0$, $d \geq 0$: rate-normalised scorers always degenerate to q_min; total-value scorers always degenerate to q_max. This is not a calibration failure — the proof holds for all parameter values — and establishes queue-global information as a necessary condition for value-aware scheduling, confirmed empirically in Section 5.23. The Value-Rate Fairness Index (VRFI = 1 − CV(VLR)) identifies value disparity that Jain's Fairness Index and slowdown variance miss: value_aware achieves zero starvation and rank sum 17 across five policies, and VRFI-targeted parameter optimization achieves the first positive VRFI (+0.289) on the Alibaba trace — while efficiency-targeted optimization degenerates through floor collapse and base shrink regardless of regularization strength.

Three open problems remain. First, the n_active = 5 failure on synthetic data: $H \approx 0.62$ at termination produces near-mean-pooling at 4 competitors, erasing the ordering information the attention was introduced to capture. The fix is a learnable inverse-temperature parameter β (scoring as $\beta/\sqrt{d}$ rather than $1/\sqrt{d}$) trained alongside the attention weights. Second, the attention diagnostic on synthetic data revealed that correct decisions are characterised by sharp focus on the longest competitor (90.6% in agree cases vs 62.4% in disagree cases): sharper attention correlates with better decisions, further motivating a learnable temperature. Third, on real traces, the area-integral reward creates gradient pressure toward longest-competitor attention that is locally coherent but globally anti-SRPT — a degeneracy that learnable temperature alone cannot resolve and that may require dense per-step reward shaping aligned with SRPT or supervised pre-training on SRPT demonstrations as a warm-start before RL fine-tuning.

Extended to real workloads via the Alibaba 2018 batch_task trace (14.3M tasks, p95/p50 = 39.7×), the agent required two preprocessing steps to become tractable: filtering tasks above the 75th percentile duration (reducing within-episode p50 burst ratio from 74× to 14×) and log- normalization of burst durations. With these changes, the W8c agent achieved mean MCT within 1.53s of Round Robin (23.30s vs 21.77s) with lower variance (10.85s vs 13.61s) on filtered test episodes — representing a 94% reduction in the Round Robin performance gap compared to the unfiltered baseline (+26.24s). The attention mechanism stabilized at H = 0.32–0.50 but learned to attend to the longest competitor with 93.7% frequency — a coherent but anti-SRPT strategy under the area-integral reward signal. The architecture is in place; the remaining obstacles are the reward signal degeneracy on real workloads and the upper quartile of the duration distribution that filtering currently excludes.

The complete real-trace progression — from 135.76s (linear normalization, full trace) to 19.22s (log normalization, filtered trace, 5-feature state) — required four interventions: outlier filtering (p75 cutoff), log-normalization, lambda_ent = 0.30 entropy regularization, and resource-class features. Each intervention was motivated by a specific diagnosed failure. The W9 agent beats Round Robin by 2.09s on mean MCT while maintaining lower variance than Round Robin (11.58s vs 14.03s), and closes the SRPT gap to 4.03s on filtered Alibaba 2018 batch_task data. The tabula rasa constraint — no expert demonstrations, no handcrafted priority rules — is preserved throughout: plan_cpu and plan_mem are raw trace fields requiring no scheduler domain knowledge to define. The complete intervention sequence after W9 tested four approaches: PBRS $\alpha = 0.20$ (F17, regressed), PBRS $\alpha = 0.05$ (F17, flat), learnable beta (F18, regressed to softer attention), and two-head attention (F19, improved). Two-head attention is the only post-W9 intervention that improved performance, reducing mean MCT from 19.22s to 18.21s and closing the SRPT gap from 4.02s to 3.01s. The improvement mechanism — spontaneous head specialization into context and monitoring roles — confirms that single-head attention was the architectural bottleneck, not the reward signal or feature set. The remaining 3.01s gap to SRPT and tier0 collapse at n_active ≥ 4 identify feature discriminability at high contention as the binding constraint, addressable only with measured runtime utilization unavailable in the Alibaba batch_task schema.

All experiments: Python 3.10+, pure NumPy + stdlib (no PyTorch / TensorFlow). UCSC Graduate Capstone, Weeks 1–10. Generated with Claude Code (claude-sonnet-4-6).